Clicks or Comments? The quality-quantity trade-off of review systems*

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I study the optimal design of review systems. A platform learns a product's quality from user-generated reviews. It faces a trade-off between the informativeness and frequency of reviews. Detailed reviews are individually more informative but less frequently submitted than simple reviews. I characterize the informational content of review systems, which depends on the information in each review and their submission rate. I use this characterization to derive the platform's optimal review system and show how it varies with reviewers' preference heterogeneity. In particular, when reviewers are homogeneous the platform isolates "horrible" experiences instead of differentiating between "good" and "bad" experiences.

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1 Introduction

Individuals and platforms often turn to others' experiences in order to learn about unfamiliar products. These experiences are frequently collected through review systems where it is infeasible to compensate agents for their information. In some cases this is due to concerns that compensation may bias the information that agents provide (e.g., Amazon prohibiting vendors from paying customers for reviews). In others, compensation is not a viable option (e.g., a school eliciting students' course reviews). In these settings, agents freely choose whether to provide their information to a platform who uses it to make choices. Since the platform lacks a way to directly incentivize agents, it must carefully decide what information to elicit. A key question is how complex elicited reviews should be. Reducing the number or complexity of questions that a reviewer must answer increases the likelihood that she will complete the review (Bean and Roszkowski 1995) at the cost of the information contained in each review. The platform must also decide how to structure its reviews. For instance, should the platform symmetrically or asymmetrically distinguish between positive and negative experiences? Different platforms have arrived at different solutions: Implemented review systems differ both in their complexity (e.g., free-text versus binary reviews) and structure (e.g., Rotten Tomatoes symmetric versus eBay's asymmetric binary reviews).

This paper models the design of review systems and quantifies these complexity and structure trade-offs. I show when a platform prefers to elicit frequent coarse information rather than scarcer detailed information. I also show how the platform optimally coarsens information. Together, these results characterize the optimal review system. In my model, a platform aims to learn the unknown quality of a product. Reviewers each have a signal of the product's quality. The platform chooses a "review system", which is a mapping from signals into reviews. Reviewers observe the review system and choose whether or not to submit a review. This choice depends on a private benefit from reviewing and the complexity of the review system as proxied by the number of possible reviews: reviewers are less likely to leave a review when they must decide between many possible options. Thus, the platform faces a trade-off between more frequent and more detailed information. The platform also faces the choice of how to structure the information that it elicits. For example, in binary review systems, the platform must choose what constitutes a “thumbs-up” or “thumbs-down” review.

Motivated by online reviews, I focus on the case when the number of reviewers is large and each reviewer is informationally small. I first leverage large deviations techniques to characterize the rate at which the platform learns the state, for each review system. This rate depends on both the average precision of a single review and the probability that a review is left. When the number of reviewers is large, the probability that the platform takes a sub-optimal action is dominated by this learning rate. As a result, regardless of the platform’s decision problem, this rate determines the platform’s preference over review systems. I then use this measure to characterize the relative information between a coarse review system and the full-detail review system (the review system defined by the identity mapping). The relative information measures how much more frequently simple reviews must be submitted for them to be preferred to the full-detail review system. When reviewers are informationally small, the relative information admits a straightforward representation, allowing me to derive the platform’s optimal review system. In particular, I show that, for any number of reviews k, the optimal review system with k reviews is a threshold system (signals are grouped in intervals). I then characterize the optimal thresholds and show how the platform optimizes over the number of reviews.

The optimal review system depends critically on reviewers’ information. To illustrate, I apply my results to study how reviewers’ heterogeneity affects the optimal binary review system.[1] I interpret reviewers’ signals as their realized utility from experiencing the product. This utility depends on the quality of the product, as well as an idiosyncratic taste component. Signals are noisy because of these idiosyncratic components: a bad experience may be due to individual taste and not the product’s quality. The distribution of these idiosyncratic preferences differ depending on the product. For example, idiosyncratic tastes for products with complex, subjective qualities (e.g., movies) often exhibit higher heterogeneity than those with simple, objective traits (e.g., toasters).

My characterization shows that the optimal binary review system is defined by

[1] The impact of taste heterogeneity on review systems has previously been studied in the empirical literature on reviews. See Price, Feick, and Higie (1989), Feick and Higie (1992), and Chu, Roh, and Park (2015).

a single threshold, above which signals are mapped to a "thumbs-up", and below which signals are mapped to a "thumbs-down". The placement of this threshold depends on reviewers' homogeneity, which I parameterize as the probability of extreme idiosyncratic tastes. When reviewers are sufficiently heterogeneous, the platform uses a symmetric threshold and asks reviewers if their experience was positive ("good") or negative ("bad"). However, when reviewers are sufficiently homogeneous the platform uses an asymmetric threshold: it is optimal to ask reviewers if their experience was "horrible" by isolating extreme negative signals (i.e., by asking if their toaster broke).

Regardless of the level of reviewers' homogeneity, if a reviewer is 3.25 times as likely to submit a binary review as a full-detail review, the optimal binary system is preferred to the full-detail review system. Performance decreases in homogeneity, but the decrease is attenuated by the careful choice of threshold. For instance, while the "good"/"bad" system must be reported only slightly more frequently than the full-detail review when reviewers are sufficiently heterogeneous (it loses only a small amount of information), its performance is arbitrarily poor when homogeneity is large. On the other hand, "horrible" review systems perform worse than the "good"/"bad" system when reviewers are heterogeneous, but bound information loss when reviewers are homogeneous. This stark difference in performance highlights the need for careful design of simple review systems. I also show that adding one review can make a large difference in performance: the optimal trinary review system is always preferred to the full-detail review system if it is reported 1.62 times as frequently. Since even binary review systems perform very well when reviewers are heterogeneous, this improvement comes from homogeneous populations and is due to the additional review isolating extreme positive signals. Together, these results suggest that even extremely simple review systems can perform very well in a variety of settings.

As this application shows, even when the platform does not know the details of reviewers' information, my results can help guide the design of review systems. In addition, these results explain several stylized facts about review systems. Many review systems are positively skewed, especially in settings when reviewers' exhibit homogeneous preferences (e.g., eBay, Uber, and Airbnb, see Hu, Zhang, and Pavlou (2009)), in contrast to review systems for products with heterogeneous preferences (e.g., Rotten Tomatoes). As well, the strong performance of binary review systems helps explain why platforms do not encourage a more uniform distribution of reviews in 5-star review systems.[2] However, the performance improvement from the binary to trinary review system suggests that platforms should reserve their review systems' highest score for "amazing" experiences.

1.1 Related Literature

This paper is closely related to the literature studying learning dynamics in recommendation and review systems. A large strand of this literature focuses on social learning dynamics. Ifrach et al. (2019) studies a standard social learning setting with binary reviews and shows when agents eventually learn the quality of the product. Besbes and Scarsini (2018) develops a similar setting, and focuses on when boundedly-rational agents' use of summary statistics of reviews leads to eventual learning. Most similar to my work is Acemoglu et al. (2022), which studies learning rates for different review systems in the context of social learning. Although both that study and this work are interested in learning rates, there are several major differences. In particular, instead of focusing on the effect of social learning, I focus on quantifying the trade-off between different review systems. Additionally, I study a setting where a key difference across review systems is the rate at which reviews are submitted. In the setting of Acemoglu et al. (2022) reviews are automatically submitted and as a result detailed review systems are always preferred to simpler systems.

A wider literature aims to study review systems outside of the context of standard social learning settings. For instance, Che and Hörner (2018) studies a platform that aims to determine the quality of a product by "spamming" it to consumers. The signal structure in that model is perfect good news, but the platform cannot prove the good news to consumers. Che, Kim, and Zhong (2024) studies how statistical discrimination can arise in markets where consumers use coarse rating schemes to choose between products. A different approach is taken in Garg and Johari (2019), which uses large deviations techniques to optimally differentiate many underlying states by controlling the proportion of positive binary reviews for each quality level. A major difference between this work and Garg and Johari (2019) is that there, the designer has full control over the information system, whereas in my case the designer

[2] It has been well-documented that most 5-star review systems are bimodal, see Chen, Yoon, and Wu (2004).

is constrained by reviewers' information.

My work is also informed by the empirical literature on online reviews. For a survey of empirical work studying review systems, see Magnani (2020). Across platforms, the distribution of submitted 5-star ratings is positively skewed and bimodal (Chen, Yoon, and Wu 2004; Hu, Zhang, and Pavlou 2009). Hu, Pavlou, and Zhang (2017) studies how this distribution is due to a self-selection effect in the population of reviewers. Fradkin, Grewal, and Holtz (2021) and Fradkin and Holtz (2023) study possible interventions to improve the rate and quality of review submission. These works find that while these interventions (including directly incentivizing reviews) can increase the rate at which reviews are submitted, they do not tend to improve the overall informativeness of the review system. This highlights the need to carefully design review systems, even when incentives can be used. The impact of heterogeneity on review systems has also been studied empirically (Price, Feick, and Higie 1989; Chu, Roh, and Park 2015). Recent empirical work has also provided alternative reasons to use coarse review systems. For instance, Botelho et al. (2025) documents that moving from a five-star system to a binary rating decreased racial discrimination in an online platform that matches workers with customers. My work complements this literature by providing an information-theoretic argument for using simple review systems, and suggesting that skewed reviews may be optimal.

My paper also contributes to the literature studying and applying rates of learning to understand economic settings. Rosenberg and Vieille (2019) and Hann-Caruthers, Martynov, and Tamuz (2018) study how quickly actions converge in a social learning setting, and Harel et al. (2021) and Dasaratha and He (2024) study the impact of networks on social learning.

At a technical level, my paper contributes to the literature on large deviations methods and comparing statistical experiments. Chernoff (1952) provides foundational results in the context of hypothesis testing. Torgersen (1981), which develops a generic ordering over different statistical experiments for finite decision problems when the number of signals is large, extends the ordering of Blackwell (1951). My characterization of the trade-off between frequency and quality in review systems extends these results to allow for stochastic reporting. Additionally, my characterization for the relative information between review systems when reviewers are informationally small is new. Large deviations techniques have been applied in several settings in the economics literature. Moscarini and Smith (2002) more sensitively studies the rate of learning, in order to determine characteristics of the demand for information. Frick, Iijima, and Ishii (2024b) ranks different misspecified models by the rate at which they slow learning.[3] Mu et al. (2021) develops a generalization of the ordering of Blackwell (1951) in the context of many signals. Finally, Fedorov, Mannino, and Zhang (2009) studies similar trade-offs to those I study in the context of hypothesis testing.

The outline of the paper is as follows. Section 2 presents and discusses the model. Section 3 formalizes the trade-off between review informativeness and reviewers' frequency of review and explicitly characterizes the optimal review system when reviewers are informationally small. Section 4 applies these results to study the impact of taste heterogeneity on review systems. Section 5 discusses several extensions and Section 6 concludes. All proofs are included in the Appendix.

2 Model

A state $\theta \in \Theta := \{L, H\}$ (e.g., the quality of a product) is drawn according to common prior $q \in (0, 1)$ that the state is L. A platform chooses an action from finite action set $\mathcal{A}$. The platform's payoff depends on its action and the state, $u : \mathcal{A} \times \Theta \to \mathbb{R}$.

There are N reviewers, indexed by $i \in \{1, ..., N\}$. Each reviewer observes a signal $S_i \in \mathbb{R}$, drawn from state-contingent density f_θ. Signals are independent conditional on the state. Let s_i denote a generic realization of the random variable S_i.

The platform chooses a review system, which is a measurable function mapping signals to reviews, $r : \mathbb{R} \to \mathbb{R}$. For instance, the platform can ask reviewers if their signal is above or below some threshold τ: $r(s_i) = \mathbb{1}\{s_i > \tau\}$. If the platform chooses review system r, and a reviewer with signal s_i submits her signal, the platform observes $r(s_i)$. Let $\mathcal{R}_r = \operatorname{supp}(r(S_i))$ denote the set of reviews possible under r.

Reviewers bear a cost to submit a review, which can be interpreted as either cognitive or temporal. The cost is $c(r) = C(|\mathcal{R}_r|)$ for some increasing function $C : \mathbb{N} \cup \{\infty\} \to \mathbb{R}_+$: it is less costly to submit reviews with fewer choices.[4] Reviewer

[3] Frick, Iijima, and Ishii (2023) and Frick, Iijima, and Ishii (2024a) apply similar techniques to different settings.

[4] Much recent work has shown that individuals have preferences against complexity (e.g., Oprea (2020)). The number of objects under consideration is a driver of complexity (Puri 2022). This has

i also receives a payoff $w_i \geq 0$ from reviewing. This benefit from reviewing reflects the expressive or altruistic desire of a reviewer to share her information and is distributed according to random variable W_i. The W_i are independent across reviewers and are independent from signals. She receives utility $w_i - c(r)$ from reviewing, and 0 utility from not. It follows that reviewers use a simple threshold rule: if $w_i \geq c(r)$ reviewer i will review, and if $w_i < c(r)$ she will not. Let $p_r := \mathbb{P}(W_i \geq c(r))$ be the ex-ante probability that a reviewer submits a review r.

The timing is as follows. First, the platform chooses a review system. Then, reviewers' private signals and benefit from reviewing are drawn, and reviewers decide whether to submit their reviews. After this, the platform observes the submitted reviews, updates its beliefs about the state, and takes an action to maximize its expected utility. Explicitly, let $I \subseteq \{1, ..., N\}$ be the set of reviewers who choose to submit a review. The platform observes $\mathcal{S}(I) = (\tilde{r}(s_1), ..., \tilde{r}(s_N))$, which is the vector of *submitted* reviews:

$$\tilde{r}(s_i) = \begin{cases} r(s_i) & \text{if} \quad i \in I, \\ \emptyset & \text{if} \quad i \notin I. \end{cases}$$

After observing $\mathcal{S}(I)$, the platform updates its belief. Let $\pi(r, \mathcal{S}) = \pi(L|r, \mathcal{S}(I))$ denote the posterior belief on state L after observing reviews $\mathcal{S}(I)$, where I drop the reliance on I for clarity. After updating its beliefs, the platform chooses an action to maximize its expected utility given beliefs $\pi(r, \mathcal{S})$:

$$u^*(r, \mathcal{S}) := \max_{a \in A} \left\{ \pi(r, \mathcal{S}) u(a, L) + (1 - \pi(r, \mathcal{S})) u(a, H) \right\}.$$

Since $\mathcal{S}$ is a random vector, $u^*(r, \mathcal{S})$ is stochastic. Define $u^*(r, N) := \mathbb{E}[u^*(r, \mathcal{S})]$, where this expectation is taken with respect to the S_i and W_i. The objective of the platform is to choose a review system to maximize $u^*(r, N)$:

$$\max_{r:\mathbb{R} \to \mathbb{R}} u^*(r, N).$$

been well-document in the case of surveys and reviews (see Bean and Roszkowski (1995)). Recently, Wang and Li (2025) studied the impact of increased cognitive load on user engagement on online platforms. That work finds that decreases in complexity lead to more user engagement.

2.1 Discussion of Model

In many contexts principals must use others' information to make informed decisions. My model provides a framework where asking simpler questions makes agents more willing to provide their information. Importantly, I do not take a stance on the goal of the platform. In some settings, the platform's goal is to simply share the information that it collects with future agents. In other settings, the platform may want to skew the information that it provides to future agents in order to encourage different behaviour (e.g., pushing a specific product). Since the model does not assume that the platform's goals are aligned with future agents, it sheds insight on a variety of settings, from online retailers to general information dissemination platforms.

This highlights a major distinction between the platform in my model and previous models of review systems. For instance, the platform's goal in designing a rating/recommendation system in Acemoglu et al. (2022) and Che and Hörner (2018) is to maximize the expected utility of consumers. The primary focus of these papers is how platforms can incentivize consumers to explore a product of unknown quality. In my work, the platform's incentives may not be aligned directly with consumers. I also abstract from the problem of experimentation in order to directly compare review systems. In practice, a platform can optimally collect information for itself while considering what is optimal to show consumers.

A key assumption is that the probability a reviewer submits a review is independent of a reviewer's signal realization. This significantly simplifies the exposition of my results, but is unrealistic: in practice reviewers with more extreme signals are more likely to submit reviews (Lafky 2014; Hu, Pavlou, and Zhang 2017). I extend the model to allow for this dependence in Section 5.1. The main insights of the baseline model extend to that setting.

One practical benefit to using simple review systems is that they are easier to digest and process than full-length reviews. I abstract from processing costs throughout. I also assume that there is no possibility of miscommunication: when a reviewer submits a review, it is reported perfectly. Abstracting from these forces allows me to focus on the key quantity-quality trade-off between simple and detailed reviews. These forces provide an additional reason for the platform to use simple reviews.

3 The Informational Content of Review Systems

In this section, I quantify the informational content of review systems when there are many reviewers. I first show how a platform trades off between quality (appropriately defined) and frequency. I then characterize the optimal review system when the reviewers' signals are imprecise, to capture environments when reviewers are informationally small.

3.1 Trading off Quality and Frequency

Given a review system r, let γ_θ^r denote the distribution of reviews conditional on state θ. That is, for any measurable subset of possible reviews $B \subseteq \mathcal{R}_r$ (where measurability is inherited from the Borel σ-algebra),

$$\gamma_\theta^r(B) := \mathbb{P}_\theta(r^{-1}(B)) = \int_{\{s:r(s)\in B\}} f_\theta(s)ds.$$

The measures γ_L^r and γ_H^r reflect the distribution of reviews *conditional* on reviews being submitted.

Review system r is informative if γ_L^r and γ_H^r are different. As the difference increases, the review systems aggregates information more quickly. The distribution of log-likelihood ratios $\log\left(\frac{d\gamma_L^r}{d\gamma_H^r}\right)$ encodes this difference. Conditional on $\theta = H$, a review is informative of the wrong state if $\log\left(\frac{d\gamma_L^r}{d\gamma_H^r}\right) \geq 0$. For any λ, from Markov's inequality,

$$\mathbb{P}_H\left(\log\left(\frac{d\gamma_L^r}{d\gamma_H^r}\right) \geq 0\right) \leq \mathbb{E}_H\left[e^{\lambda \log\left(\frac{d\gamma_L^r}{d\gamma_H^r}\right)}\right] = \mathbb{E}_H\left[\left(\frac{d\gamma_L^r}{d\gamma_H^r}\right)^\lambda\right].$$

Hence, the moment generating function of log-likelihood ratios provides a bound on the probability of reviews that are indicative of the wrong state. Since this relationship holds for any $\lambda \in \mathbb{R}$, the minimum of the moment generating function provides the tightest bound. When the number of reviews is large, this bound is tight.[5] Hence, the minimum of this moment generating function determines the rate at which the

[5] For any informative review system, the conditional expected value of the log-likelihood ratio is negative. When the number of reviews is large, the average belief drifts towards the true state. This means that "on average", any informative review system will be correct. In turn, it is the probability of unlikely beliefs that determines the relative performance of review systems. This is why the Kullback-Leibler divergence is not the correct notion of difference in this setting.

platform ceases to make mistakes about the state. If this minimum is small, it is more likely that the platform is correct in its belief after observing reviews.

Definition 1. The *learning efficiency* of review system r is the following measure of distance between γ_H^r and γ_L^r:

$$\nu(r) := 1 - \min_{\lambda \in [0,1]} \int_{\mathcal{R}_r} \left(\frac{d\gamma_L^r}{d\gamma_H^r} \right)^{\lambda} d\gamma_H^r \in [0, 1].$$

The learning efficiency is a simple transformation of the minimum of the moment-generating function of log-likelihoods and encodes how quickly the platform learns the state. The learning efficiency of a review system r is larger if r separates the states more effectively. If $d\gamma_L^r > 0 \iff d\gamma_H^r = 0$, learning occurs immediately and $\nu(r) = 1$. If reviews are uninformative ($\gamma_L^r = \gamma_H^r$), $\nu(r) = 0$ and learning never occurs.

The speed at which the platform learns the state controls the platform's expected utility when the number of reviewers is large. Consider first the case that all reviews are submitted ($p_r = 1$). The platform's utility is determined by the probability that it takes a sub-optimal action. Since sub-optimal actions are caused by incorrect beliefs, $\nu(r)$ determines this probability. As a result, *regardless* of the platform's utility function u, $\nu(r)$ determines how quickly its expected utility converges its the full-information utility. Hence, if $\nu(r) > \nu(r')$, for large N the platform's expected utility is higher under r than under r'.[6]

Decreasing the probability that reviews are submitted p_r slows the learning of the platform. Intuitively, if reviews are submitted half of the time, the platform learns half as quickly. Theorem 1 formalizes this intuition: regardless of the decision problem, $p_r\nu(r)$ determines the utility of the platform for large N. In particular, the platform separably trades off between the quality of reviews and the rate at which they are submitted.

Theorem 1. *Fix two review systems r, r' such that $p_r\nu(r) > p_{r'}\nu(r')$. For any finite action set $\mathcal{A}$ and utility function u such that the platform's decision problem is not trivial (*$\operatorname{argmax}_{a\in\mathcal{A}} u(a, L) \cap \operatorname{argmax}_{a\in\mathcal{A}} u(a, H) = \emptyset$*), there exists an $\overline{N}$ such that for all $N \geq \overline{N}$, such that $u^*(r, N) > u^*(r', N)$.*

[6]This discussion summarizes a known result in the literature (Chernoff 1952; Torgersen 1981).

The proof of Theorem 1 leverages large deviations techniques developed for the case that $p_r = 1$. To utilize those results, I construct a statistical experiment with full reporting whose learning efficiency is $p_r\nu(r)$ and show that this experiment is equivalent to the review system with stochastic reporting. The subtlety of the proof is in the definition of $\nu(r)$. This measure admits the separability between reporting rates and learning efficiency that is not present in previous studies.

Theorem 1 implies that the platform need not tailor its review design to its decision problem because $p_r\nu(r)$ is independent of its problem. This is particularly relevant when the platform is designing a review system that is to be used across a number of settings, as is the case for large organizations.

In practice, review frequencies p_r are simple for platforms to empirically compute through A/B testing. This suggests a focus on the comparison of learning efficiencies $\nu(r)$ for different review systems r. Theorem 1 is equivalent to: if

$$\frac{\nu(r)}{\nu(r')} > \frac{p_{r'}}{p_r}, \tag{1}$$

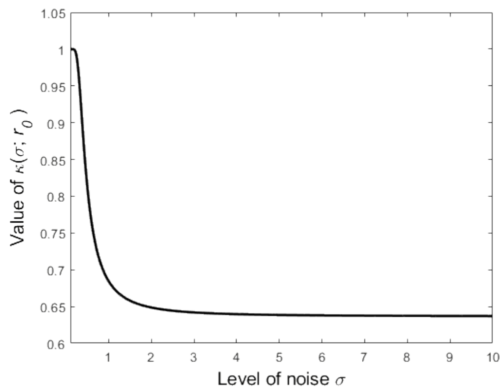
then there exists an $\overline{N}$ such that for all $N \geq \overline{N}$, such that $u^*(r, N) > u^*(r', N)$. Heuristically, if r' is submitted much less frequently than r ($p_{r'} \ll p_r$), then the platform prefers r. The ratio $\nu(r)/\nu(r')$ formalizes how small $\frac{p_{r'}}{p_r}$ must be for r to be favoured. Consider in particular the full-detail review $r_f(s) = s$. By setting $r' = r_f$, the value (1) becomes a measure of how much information a review from review system r contains relative to the full-detail review r_f. I next define a measure of relative information in order to compare review systems to the full-detail review system.

Definition 2. The *relative information* of a review function r is $\kappa(r) := \frac{\nu(r)}{\nu(r_f)} \in [0, 1]$.

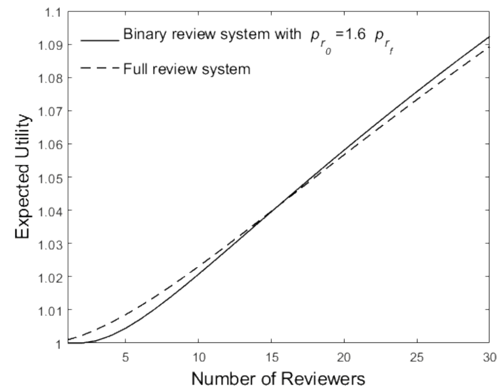
No review system can contain more information in an individual review than the full-detail review system. The relative information $\kappa(r)$ of a review system r measures how much information is lost when coarsening from the full-detail review to r. $\kappa(r)$ characterizes how much less frequently the full-detail review r_f must be submitted than r for the platform to prefer eliciting reviews of type r to full-detail reviews. For instance, if $\kappa(r) = 0.5$, then r must be submitted twice as often as the full-detail review for the platform to prefer r. The relative information compares the informational content across review systems in terms of how frequently they are reported. To directly compare two different review systems r and r', the platform uses $\kappa(r)/\kappa(r')$. In the following example I illustrate how Theorem 1 and the relative information can be used in practice.

Example 1. *Suppose that $q = 0.5$ and reviewers' signals are given by $s_i|(\theta = L) \sim N(-1, \sigma^2)$ and $s_i|(\theta = H) \sim N(1, \sigma^2)$. The platform is deciding between implementing the full-detail review system ($r_f(s) := s$) and asking reviewers if their signal is positive or negative ($r_0(s) := \mathbb{1}\{s \leq 0\}$). It can be shown that $\nu(r_f) = 1 - e^{-\frac{1}{2\sigma^2}}$ and $\nu(r_0) = 1 - 2\sqrt{\Phi\left(\frac{1}{\sigma}\right)\left(1 - \Phi\left(\frac{1}{\sigma}\right)\right)}$, where Φ is the cdf of the standard normal distribution. The relative information $\kappa(r_0)$ for this review system is plotted as a function of σ in Fig. 1a. As this figure shows, for any degree of noise, relative information of the binary review system $\kappa(r_0)$ is at least $\frac{1}{1.57}$. This means that, as long as the binary review is reported 57% more frequently than the full-detail review, it is preferred by the platform when the number of reviewers is sufficiently large. The figure also shows that in this example, when reviewers' signals are more precise the binary review system performs better: $\kappa(r_0)$ is decreasing in σ. For example, when $\sigma = 0.5$ then binary reviews must be reported only 23% more frequently; when $\sigma = 1$ they must be reported 46% more frequently.*

Theorem 1 shows that if $p_{r_0} > p_{r_f}/\kappa(r_0)$, the binary review system is preferred when the number of reviewers is large. While the exact value of N depends on the decision problem, for simple decision problems this asymptotic bound applies quickly. Suppose that the platform cares about guessing the quality of the product, but cares more about determining when the product is of low quality: $A = \{L, H\}$ and $u(a, \theta) = \mathbb{1}\{a, \theta = H\} + 2 \cdot \mathbb{1}\{a, \theta = L\}$. Fig. 1b plots $u^*(r_f, N)$ and $u^*(r_0, N)$ as a function of the number of reviewers N for $\sigma = 5$, $p_{r_f} = 0.2$. Here, $\frac{1}{\kappa(r_0)} \approx 1.57$. Setting $p_{r_0} = 1.6 \cdot p_{r_f}$, so that the binary review system is just asymptotically preferred, shows how quickly this asymptotic preference applies. When the number of reviewers N is small, the full-detail review system is preferred, because of the small chance of very informative signals. However, this gap quickly disappears as N increases. Even when the expected number of full-detail reviews is 4 ($N = 20$), the binary review system r_0 is preferred.



(a) Relative information of the binary review system r_0, as a function of noise σ.



(b) Values of $u^*(r_f, N)$ and $u^*(r_{r_0}, N)$, for $\sigma = 5$. Here $p_{r_f} = 0.2$ and $p_{r_0} = 1.6 \cdot p_{r_f} \gtrsim p_{r_f}/\kappa(r_0)$.

Figure 1

Remark 1. It is important for the separability of $\nu(r)$ and p_r that the reporting rate p_r is independent of a reviewer's realized signal. As mentioned previously, I extend the model to allow an individual reviewer's reporting rate to depend on her realized signal in Section 5.1.

Remark 2. The p_r represent the stochastic rate at which signals are reported. An alternative is to ask how many signals n_r from r "equals" $n_{r'}$ signals from r'. I discuss this alternative in Section 5.2.

3.2 Threshold Systems and Imprecise Reviews

An important class of review systems is the class of *threshold systems*. In practice, review systems are monotone: a more negative experience leads to a more negative review. Threshold systems capture this behaviour. In this section I show that threshold systems are optimal for a wide class of signal structures and characterize the optimal k-threshold system when reviewers' signals are imprecise.

Definition 3. A review system r is a k-threshold review system if there exist $\tau_0 = -\infty < \tau_1 < ... < \infty = \tau_k$ such that

$$r(s) = \sum_{i=1}^{k} r^i \cdot \mathbb{1}\left\{s \in (\tau_{i-1}, \tau_i]\right\}, \text{ for distinct } r^i \in \mathbb{R}.$$

I write $r_{\boldsymbol{\tau}}$ for the review system defined by thresholds $\boldsymbol{\tau} = (\tau_1, ..., \tau_{k-1})$.

Threshold systems are induced by natural partitions of the signal space $\mathbb{R}$: all signals contained in the interval $(\tau_{i-1}, \tau_i]$ are mapped to the same review r^i. In order to study threshold review systems, I parameterize the precision of an individual review by assuming that a reviewer's signal is given by $S_i = \mu_\theta + \sigma\epsilon_i$, where

$$\mu_\theta = \begin{cases} -\mu & \text{if} \quad \theta = L, \\ \mu & \text{if} \quad \theta = H. \end{cases}$$

A reviewer's signal can be interpreted as her perceived realized utility from a product whose quality is dependent on the state. Under this parameterization the μ_θ reflect the average utility induced by the good. The ϵ_i reflect the idiosyncratic taste component in reviewers' experiences of the good: different reviewers experience the same product differently. The parameter σ controls the salience of the idiosyncratic taste shocks. When σ is large, idiosyncratic tastes overwhelm the average utility differences across states. This corresponds to setting when reviewers are informationally small. For this reason I refer to σ as the level of noise in reviewers' signals. Let f be the density of ϵ_i. In this setting, f_L and f_H are given by

$$f_L(s) = \frac{1}{\sigma} f\left(\frac{s+\mu}{\sigma}\right) \quad \text{and} \quad f_H(s) = \frac{1}{\sigma} f\left(\frac{s-\mu}{\sigma}\right).$$

The following is a regularity assumption for f to guarantee that the platform's problem is well-behaved in this setting.

Assumption 1. The density of idiosyncratic taste shocks f has full support and is continuous. For all $\lambda \in [0, 1]$ there exists an $\epsilon > 0$ such that, for all $\mu \in (0, \epsilon)$ $f(s+\mu)^\lambda f(s-\mu)^{1-\lambda}$ is twice differentiable with respect to μ almost everywhere, and these derivatives are integrable with respect to s.

Review systems lose information because they combine signals that induce different beliefs. When a single review combines signals that induce very different beliefs, information loss is large. Hence, it is always in the platform's interest to group signals that induce similar beliefs. When signals are ordered in terms of informativeness, threshold systems are optimal because signals that induce similar beliefs are "close". Log-concavity of f ensures that lower signals are more informative of $\theta = L$, guaranteeing that threshold systems are optimal.

Lemma 1. *Suppose that f is log-concave. For any $k \in \mathbb{N}$ and any level of noise $\sigma > 0$, the optimal review system with k reviews is a k-threshold system.*

Given the optimality of threshold systems, I now seek to characterize the optimal threshold review system. A key feature of online reviews is that a single review is imprecise: the level of noise σ is large. Optimizing the rate of learning is especially relevant in this setting since small sample effects have a minor impact.

In order to simplify notation, I transform signals: $s \mapsto \frac{s}{\sigma}$. This transformation preserves the information at a fixed σ while normalizing the scale of signals, guaranteeing that review systems converge. In what follows I deal exclusively with normalized signals. The distributions of normalized signals in the two states are given by

$$f_L(s) = f\left(s + \frac{\mu}{\sigma}\right) \quad \text{and} \quad f_H(s) = f\left(s - \frac{\mu}{\sigma}\right).$$

Given a review system r, write the learning efficiency of r when the level of noise is σ as $\nu_f(\sigma; r)$. Similarly, write the relative information of r as $\kappa_f(\sigma; r)$. When it is clear from the context, I drop the dependence on f. As the noise in individual signals σ increases, the distributions of signals in the two states become similar, limiting the information contained in an individual review. In the limit ($\sigma \to \infty$), individual reviews contain no information ($\lim_{\sigma\to\infty} \nu(\sigma; r) = 0$ for all r). However, the relative information of reviews remains well-defined. In order to characterize the optimal review system, I explicitly characterize $\lim_{\sigma\to\infty} \kappa(\sigma; r_{\boldsymbol{\tau}})$ for threshold systems $r_{\boldsymbol{\tau}}$. Slightly abusing notation, I write $\kappa(\infty; r_{\boldsymbol{\tau}})$ for this limit:

$$\kappa(\infty; r_{\boldsymbol{\tau}}) := \lim_{\sigma\to\infty} \kappa(\sigma; r_{\boldsymbol{\tau}}) = \lim_{\sigma\to\infty} \frac{\nu(\sigma; r_{\boldsymbol{\tau}})}{\nu(\sigma; r_f)}.$$

If $\kappa(\infty, r_{\boldsymbol{\tau}})$ is small, then reviews from $r_{\boldsymbol{\tau}}$ must be reported much more frequently than full-detail reviews in order for the platform to prefer $r_{\boldsymbol{\tau}}$ when reviewers signals are imprecise. On the other hand, if $\kappa(\infty, r_{\boldsymbol{\tau}})$ is large, then even when reviewers are slightly more likely to leave the simpler review $r_{\boldsymbol{\tau}}$, the platform prefers eliciting coarse information.

Before I introduce the result, some intuition is useful. When noise is large, on regions where f is nearly constant, signals are uninformative. This is because the difference between the two densities f_L and f_H is small. However, on regions where f is very curved, even when noise is large, signals are very informative. If f is very curved, the full-detail review system remains informative even when signals are imprecise. In turn, a large curvature in f depresses $\kappa(\infty, r_{\boldsymbol{\tau}})$. The measure of curvature that captures this effect is the *Fisher Information* of f:[7]

$$I_f := \mathbb{E}\left[\left(\frac{f'(s)}{f(s)}\right)^2\right].$$

A similar logic determines the behaviour of $\nu(\sigma; r_{\boldsymbol{\tau}})$ when σ is large. Since the review system is discrete, the effect of a marginal change in σ only impacts reviews at the thresholds $\tau_1, ..., \tau_{k-1}$. Consider a single review $r^i \in \mathcal{R}_{r_{\boldsymbol{\tau}}}$. A marginal decrease in noise increases the probability that r^i is reported in state L on the order of $(f(\tau_i) - f(\tau_{i-1}))$. Symmetrically, the probability that r^i is reported in state H decreases by the same amount. Review r^i becomes more informative as noise decreases if the magnitude of this difference is large. This effect is magnified if r^i is uncommon because the difference in probabilities in the two states is more easily detected.

Assumption 1 guarantees that the Fisher Information is well-defined. These effects come together to characterize relative information when signals are noisy:

Lemma 2. *Assume that f satisfies Assumption 1. Let F denote the cumulative distribution function of f. Then, the relative information of review system $r_{\boldsymbol{\tau}}$ with thresholds defined by $\boldsymbol{\tau} = (\tau_1, ..., \tau_{k-1})$ when information is imprecise is given by*

$$\kappa(\infty; r_{\boldsymbol{\tau}}) = \left(\sum_{i=1}^{k} \frac{(f(\tau_i) - f(\tau_{i-1}))^2}{F(\tau_i) - F(\tau_{i-1})}\right) \Big/ I_f \in [0, 1],$$

with the convention that $f(\infty) = f(-\infty) = 0$, and $F(\infty) = 1, F(-\infty) = 0$.

Lemma 2 computes the performance of threshold systems when reviewers' signals are imprecise.[8] It is proved by showing independently the rate at which $\nu(\sigma; r_f)$ and $\nu(\sigma; r_{\boldsymbol{\tau}})$ go to zero as a function of the noise in the taste idiosyncrasies. Lemmas 1 and 2 together characterize the optimal review system with k reviews.

[7]The Fisher information appears in other settings with noisy signals. For instance, in Sadzik and Stacchetti (2015), a larger Fisher information corresponds to easier detection of defections in a principal-agent model with frequent, but noisy, signals.

[8]Lemma 2 can also be extended to non-threshold finite systems, see Theorem 3.

Theorem 2. *Assume that f is log-concave and satisfies Assumption 1. The optimal review system with k reviews when information is imprecise is the k-threshold system $r_{\boldsymbol{\tau}}$ with thresholds $\boldsymbol{\tau} = (\tau_1, ..., \tau_{k-1})$ that maximize*

$$\sum_{i=1}^{k} \frac{(f(\tau_i) - f(\tau_{i-1}))^2}{F(\tau_i) - F(\tau_{i-1})}. \tag{2}$$

When signals are imprecise, this system is preferred to full-detail review system r_f if

$$\frac{\sum_{i=1}^{k} \frac{(f(\tau_i)-f(\tau_{i-1}))^2}{F(\tau_i)-F(\tau_{i-1})}}{I_f} > \frac{p_{r_f}}{p_{r_{\boldsymbol{\tau}}}}. \tag{3}$$

The full-detail review system is preferred to $r_{\boldsymbol{\tau}}$ if the reverse inequality holds.

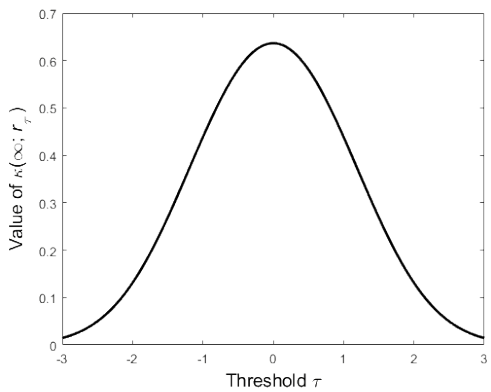
Theorem 2 provides a straightforward objective for the platform to maximize.[9] Moreover, by characterizing the optimal review system with k reviews, Theorem 2 implicitly characterizes the optimal review system. To determine the optimal review system, the platform first optimizes for a fixed number of reviews and then optimizes over the number of reviews. Theorem 2 also shows that, even when noise is large and reviewers are informationally small, the choice of thresholds has important implications for the relative information of review systems. The following example shows how to choose the optimal review system in practice.

Example 2. *Consider the setting of Example 1. Now, instead of deciding between the full-detail review system r_f and the positive/negative binary review system r_0, the platform wishes to implement the optimal review system. The platform first designs the optimal review system for each possible number of reviews k. To do this, for each $k \in \mathbb{N}$, it chooses the $k-1$ thresholds that maximize (2), yielding the optimal k-review system $r_{\boldsymbol{\tau}_k}$. Fig. 2a plots the objective in the case of $k = 2$: here, the review system with threshold zero r_0 is the optimal binary review system.*

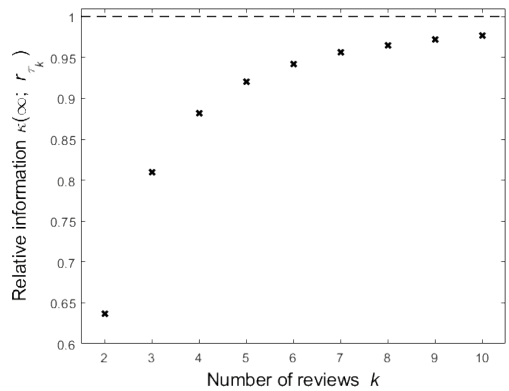
The platform next optimizes the number of reviews k by comparing $r_{\boldsymbol{\tau}_k}$. The optimal k maximizes $p_{r_{\boldsymbol{\tau}_k}} \cdot \kappa(\infty; r_{\boldsymbol{\tau}_k})$. The values of $\kappa(\infty; r_{\boldsymbol{\tau}_k})$ are plotted in Fig. 2b. Here, the informational gain from adding an additional review is large when increasing

[9] The optimal $\boldsymbol{\tau}^*$ depends on σ. However, it is sufficient to set $\boldsymbol{\tau}^* = \lim_{\sigma\to\infty} \boldsymbol{\tau}^*(\sigma)$ as $\lim_{\sigma\to\infty} \frac{\nu(\sigma; r_{\boldsymbol{\tau}^*(\sigma)})}{\nu(\sigma; r_{\boldsymbol{\tau}^*})} = 1$ under Assumption 1.

from a binary to a trinary system and from a trinary to 4-review system. The gains decrease as additional reviews are added. At 5 reviews, very little information is lost relative to the full-detail review system ($\kappa(\infty; r_{\boldsymbol{\tau}_5}) > 0.92$). Provided that the reporting rate for 5 reviews is at least 9% higher than the reporting rate for the full-detail review system, it is preferred. Thus, it is likely that the optimal review system has a small number of reviews.



(a) Relative information $\kappa(\infty; r_\tau)$ for different binary review systems r_τ.



(b) Optimal relative information $\kappa(\infty; r_{\boldsymbol{\tau}_k})$, for different numbers of reviews k.

Figure 2

4 Application: Taste Heterogeneity and Binary Review Systems

Theorem 2 shows that even when reviewers' signals are imprecise, the choice of review system impacts the platform's learning. While the level of noise σ controls the degree to which idiosyncratic tastes mask differences due to the state, it does not control the distribution of taste shocks (i.e., the distribution of ϵ_i). This distribution varies across settings and has important implications for optimal review systems. In this section I parameterize the distribution of taste shocks by reviewers' taste heterogeneity. I use this parameterization to show how the optimal binary review system varies as a function of this heterogeneity.

When reviewers are heterogeneous, they frequently exhibit extreme idiosyncratic tastes. That is, heterogeneity corresponds to large tails of f.[10] For some products

[10]Importantly, by heterogeneity I do not mean the variance of reviewers' preferences: this effect

like movies, it is common to for reviewers to exhibit extreme preferences: realized individual tastes that are multiple standard deviations from the average are frequent. However, for other products (e.g., toasters), extreme preferences are very uncommon: it is extremely unlikely that a reviewer likes a toaster two standard deviations more than the average reviewer.[11] The heterogeneity in reviewers' taste idiosyncrasies has important implications for optimal review systems.

4.1 Symmetric Heterogeneity

In order to understand how taste heterogeneity affects review systems, I restrict attention to a one-parameter family of distributions whose parameter controls the degree of taste heterogeneity within the population of reviewers:

$$f_\alpha(s) \propto \exp(-|s|^\alpha).$$

By controlling the tails of the distribution f_α, the parameter α controls the level of heterogeneity in reviewers' idiosyncratic tastes. When α is large, f_α has small tails, so that the population of reviewers is homogeneous. Henceforth, I refer to α as homogeneity.

Remark 3. For $\alpha \geq 1$, $f_\alpha(s)$ is log-concave; threshold systems are optimal. This family is referred to as the Generalized Normal Distributions and incorporates three well-known distributions as special cases: f_1 is the Laplace distribution (large heterogeneity); f_2 is the Normal distribution (moderate homogeneity); and as $\alpha \to \infty$, f_α converges almost everywhere to the uniform distribution on $(-1, 1)$ (perfect homogeneity).

I first show how relative information varies with homogeneity when the platform uses r_0 (i.e., it asks reviewers if their experience was positive or negative). When tastes are very heterogeneous, r_0 performs well: relative information is close to 1.

is absorbed by the imprecision in reviewers' signals. Instead, this heterogeneity measures how likely a signal multiple standard deviations from the mean is, holding fixed the standard deviation.

[11]The classification of "search" and "experience" goods provides a possible way to distinguish between goods with low and high taste heterogeneity. First introduced in Nelson (1970), search goods' attributes are well-defined and easily found, whereas experience goods must be experienced before an opinion can be formed (see Magnani (2020) for a discussion). Search goods, because of their well-defined attributes, are likely to be more homogeneous—"does the good perform as it should?"—while experience goods are likely to be more heterogeneous since individual experience important.

However, as homogeneity grows, relative information decreases to zero: information loss is unbounded.

Proposition 1. *Consider relative information under the threshold 0,* $\kappa_{f_\alpha}(\infty; r_0)$*:*

(i) *When reviewers are very heterogeneous, the symmetric binary review system contains the same information as the full-detail review system:* $\kappa_{f_1}(\infty; r_0) = 1$*; and*

(ii) *As reviewer taste homogeneity increases,* r_0 *performs worse relative to the full-detail review system:* $\kappa_{f_\alpha}(\infty; r_0)$ *is decreasing in* $\alpha \geq 1$*. Moreover,* r_0 *performs arbitrarily poorly when reviewers are sufficiently homogeneous:* $\lim_{\alpha\to\infty} \kappa_{f_\alpha}(\infty; r_0) = 0$.

As reviewer taste homogeneity increases, reviewers must submit binary reviews more frequently for r_0 to continue to be preferred to the full-detail review. The intuition is as follows. When reviewers are homogeneous, signals are concentrated. This means that there are many uninformative signals. However, because tails are small, differences in the state are noticeable at extreme signals. For instance, a broken toaster is much more likely if the quality of the toaster is low, and is very informative. This means that there are also some very informative signals.[12] The symmetric review r_0 mixes very informative signals with very uninformative signals, limiting the information contained in either review (i.e., it mixes signals from broken toasters with signals from reviewers who were only mildly inconvenienced).

This discussion highlights the central trade-off in choosing a binary review system: extracting more information from one review while extracting less from the other. If the platform uses review system r_τ for $\tau < 0$ instead of r_0, then (i) the positive review contains a mass of negative signals; but (ii) the negative review contains a smaller mass of uninformative signals. Effect (i) makes the review system less informative, while (ii) improves precision. If the second effect outweighs the first, then it is optimal to use an asymmetric review.

When reviewers' tastes are sufficiently heterogeneous, the cost of decreasing the informativeness of one review is larger than the benefit from making the other review more informative. However, when reviewer homogeneity is large, enough information

[12] I.e., the variance of induced posteriors is large.

is extracted from the informative review to outweigh the cost: the optimal binary review system is asymmetric. Importantly, the optimal threshold limits learning loss under the binary review system: relative information is bounded below by 1/3.25. That is, if reviewers are 3.25 times as likely to submit a binary review than the full-detail review, the optimal binary system is preferred, regardless of the degree of homogeneity. This result, together with Proposition 1, highlights the importance of this analysis: the choice of τ has important implications for a binary review system's performance.

In what follows, let $\tau^*(\alpha)$ be the non-negative optimal threshold. Since f_α is symmetric, there is also a non-positive maximizer $-\tau^*(\alpha)$. In the statement of Proposition 2, all statements are written in terms of the non-negative optimal threshold for clarity. Comparable statements hold for the non-positive optimal threshold.[13]

Proposition 2. *For $\alpha \geq 1$, there is a unique optimal binary review system with non-negative threshold $\tau^*(\alpha)$. It has the following properties:*

(i) The symmetric binary review system is the optimal binary review system if and only if reviewers are sufficiently heterogeneous: if $\alpha \leq 2$, $\tau^*(\alpha) = 0$, while if $\alpha > 2$, $\tau^*(\alpha) \neq 0$;

(ii) When reviewers are sufficiently homogeneous, the optimal binary review system is very asymmetric: $\lim_{\alpha \to \infty} \tau^*(\alpha) = 1$; and

(iii) The relative information of the optimal binary review system is bounded: $\kappa_{f_\alpha}(\infty; r_{\tau^*(\alpha)}) > 1/3.25$ for all α.

The normal distribution corresponds to the degree of taste homogeneity at which the optimal binary review system ceases to be symmetric. When the degree of homogeneity is large the platform optimally asks individuals either (i) whether they had an "amazing" (positive τ^*), or (ii) whether they had a "horrible" experience (negative τ^*).

Proposition 3 shows comparative statics for $\tau^*(\alpha)$: both $\tau^*(\alpha)$ and $F_\alpha(\tau^*(\alpha))$ are increasing in α. Importantly, this means that the probability of the more common

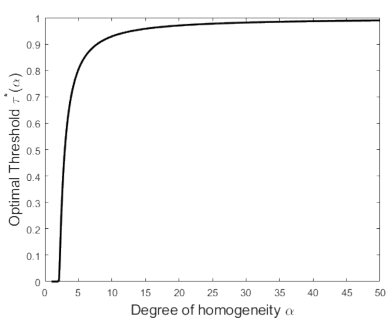
[13] The next section shows that introducing (arbitrarily small) asymmetry in f_α breaks this multiplicity.

review increases in homogeneity, *regardless of the state.* Consider $r_{\tau^*(\alpha)}$, the optimal binary review with a positive threshold, for large α. Even when $\theta = H$, the majority of reviews are negative. The share of negative reviews increases to 1 as $\alpha \to \infty$. This suggests that review systems where the vast majority of reviewers leave the same review are optimal when reviewers are sufficiently homogeneous. These results are summarized in Fig. 3. In particular, even for relatively small degrees of homogeneity, the optimal review system is very asymmetric.

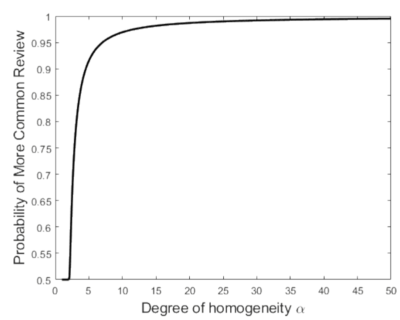
Proposition 3. *Consider the non-negative optimal threshold* $\tau^*(\alpha)$. *It has the following properties:*

(i) The optimal threshold is increasing in the homogeneity of the population: $\tau^*(\alpha)$ *is strictly increasing in* α *for* $\alpha \geq 2$*; and*

(ii) Optimal asymmetry is increasing in the homogeneity of the population: $F_\alpha(\tau^*(\alpha))$ *is strictly increasing in* α *for* $\alpha \geq 2$*, with* $\lim_{\alpha\to\infty} F_\alpha(\tau^*(\alpha)) = 1$.



(a) As homogeneity increases, the optimal review becomes more asymmetric.



(b) As homogeneity increases, the probability of the more common review increases to 1.

Figure 3: Properties of the optimal binary review system, for different degrees of homogeneity α.

4.2 Asymmetric Heterogeneity

The simultaneous optimality of the “amazing” and the “horrible” review systems disappears when asymmetry in positive and negative taste homogeneity is introduced. When heterogeneity is asymmetric, the optimal binary review system isolates extreme experiences on the more homogeneous side of the distribution.

Proposition 4. *Define α^* to be the value for which $f_\alpha(0)$ is maximized.*[14] *Let*

$$f_{\alpha_L,\alpha_H}(s) \propto \mathbb{1}\{s \geq 0\}e^{-|s|^{\alpha_H}} + \mathbb{1}\{s < 0\}e^{-|s|^{\alpha_L}},$$

with $\alpha^* \leq \alpha_L, \alpha_H$. If $\alpha_L > \alpha_H$, then the optimal threshold $\tau^*(\alpha_L, \alpha_H)$ is negative. If $\alpha_H > \alpha_L$, then the optimal threshold is positive.

Fig. 4 highlights the intuition behind Proposition 4. Consider a positive threshold (dotted line). At the negative threshold that yields the same value of the density (dashed line), there is less mass to the left of that threshold than there is to the right of the positive candidate. This logic applies to any feasible positive candidate, so that the optimal threshold is negative.[15]

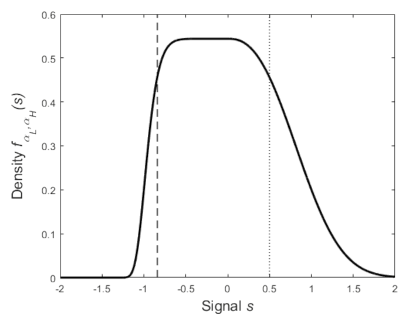


Figure 4: Asymmetric distribution of taste shocks f_{α_L,α_H}, for $\alpha_L = 10, \alpha_H = 2.5$.

Apart from breaking multiplicity, this is an important setting because asymmetric taste homogeneity is common. For instance, consider ride-sharing services. In this setting, reviewers agree on bad experiences: no reviewer wants to be late or get into an accident. However, reviewers disagree on what makes a good experience: some (not all) like having a discussion with their driver; some (not all) like music to be played.[16] Proposition 4 justifies why many platforms exhibit many more positive reviews than negative reviews (Hu, Zhang, and Pavlou 2009). If reviewers agree on negative experiences more than positive experiences, the platform *should* isolate extremely negative experiences. This means that in many natural settings reviews

[14]It is easy to verify that $\alpha^* \approx 2.166$.

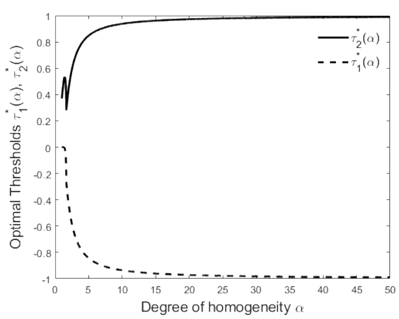
[15]If the variance of signals for positive taste shocks was larger than for negative taste shocks, then by a similar logic the optimal threshold is negative.

[16]For search goods, larger negative homogeneity is expected: most reviewers are in agreement about whether a good doesn't meet expectations, but may disagree about how it over-performs.

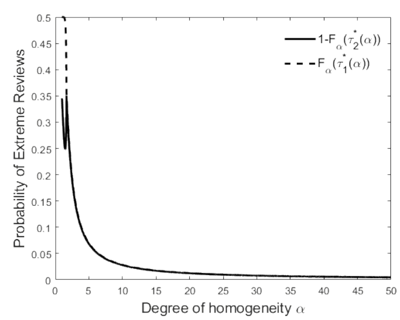
are optimally positively skewed, even when the product is of low quality.

4.3 Beyond Binary Reviews

Similar qualitative results apply to optimal finite review systems with k components, where $k > 2$. Fig. 5 exhibits the optimal three-bin system for varying degrees of homogeneity α. In particular, the optimal asymmetry of the system is increasing in the degree of homogeneity. An important difference from the binary system is that both extreme positive and negative signals can be isolated. When homogeneity is large, the platform "throws out" many signals. The middle review provides no information to the platform (since it is equally likely in each state). Its presence is important, however, because it allows the platform to skim uninformative reviews in order to isolate very informative positive and negative signals. When homogeneity is large, the optimal three-bin system performs twice as well as the optimal binary review system: it is always preferred to the full-detail review if it is submitted more than 1.62 times as frequently.



(a) The optimal thresholds become more extreme as homogeneity grows.



(b) The probability of the highest and lowest review shrinks as homogeneity grows.

Figure 5: Properties of the optimal three-bin review system, for different degrees of homogeneity α.

5 Extensions

5.1 Signal-Dependent Reporting Rates

In practice, a reviewer's experience is likely to affect the probability that she submits a review. For example, a reviewer with an extreme signal is more likely to leave a review.[17] In some contexts reviewers may report positive and negative experiences at different rates. In this section I show how to incorporate signal-dependent reporting rates into my framework. While the qualitative insights from previous sections remain, several new forces emerge.

Suppose now, in addition to the cost of submitting the review $c(r)$, there is a benefit $b(s)$ of reporting signal s (with $b(s) \leq c(r)$). This reflects an expressive component to submitting reviews that depends on the reviewer's signal. A reviewer now receives utility $w_i + b(s) - c(r)$ from submitting review r when her signal is s. In this setting, the probability that a reviewer leaves a review is a function of both the review system and her signal: $p(r, s) := \mathbb{P}(W_i \geq c(r) - b(s))$.

As before, the difference in the distribution of reviews across the two states determines the review system's performance. Define the distribution $\gamma_L^{(p,r)}$ of reviews in state L (implicitly dependent on σ) as

$$\gamma_L^{(p,r)}(B) := \int_{\{s: r(s) \in B\}} p(r, s) f\left(s + \frac{\mu}{\sigma}\right) ds.$$

Similarly define the distribution $\gamma_H^{(p,r)}$ of reviews in state H. As before, these measures determine the rate at which the platform learns the state. Importantly, $p(r, s)$ is no longer separable from these measures. Define $\rho(\sigma; r, p)$ as the analog of $p_r \nu(\sigma; r)$:

$$\rho(\sigma; r, p) := 1 - \min_{\lambda \in [0,1]} \left[\int_{\mathcal{R}_r} \left(\frac{d\gamma_L^{(p,r)}}{d\gamma_H^{(p,r)}} \right)^{\lambda} d\gamma_H^{(p,r)} + \left(\gamma_L^{(1-p,r)}(\mathbb{R})\right)^{\lambda} \left(\gamma_H^{(1-p,r)}(\mathbb{R})\right)^{1-\lambda} \right].$$

There are two significant differences between $\rho(r, p)$ and $p_r \nu(r)$. First, different signals receive different weight because $p(r, \cdot)$ varies with the signal realization. Second, "null reviews" (reviews that are not submitted) may now be informative of the state. The probability that a signal is not submitted when the state is L is $\gamma_L^{(1-p,r)}(\mathbb{R})$. If this is larger than $\gamma_H^{(1-p,r)}(\mathbb{R})$, then a null signal is indicative of state L. Despite these changes, Theorem 1 generalizes to this setting, where $\rho(\sigma; r, p)$ replaces $p_r \nu(\sigma; r)$.

[17] For instance, Lafky (2014) suggests that individuals are more likely to report extreme experiences than moderate experiences because they derive a benefit from informing others, and extreme experiences are more informative.

Lemma 3. *Fix two review systems r, r' such that $\rho(\sigma; r, p) > \rho(\sigma; r', p)$. For any finite action set $\mathcal{A}$ and utility function u such that the platform's decision problem is not trivial (*$\operatorname{argmax}_{a \in \mathcal{A}} u(a, L) \cap \operatorname{argmax}_{a \in \mathcal{A}} u(a, H) = \emptyset$*), there exists an $\overline{N}$ such that for all $N \geq \overline{N}$, such that $u^*(r, N) > u^*(r', N)$.*

Without additional structure on the probabilities of review p, little more can be said about the comparison between review systems. With arbitrary reporting rates p, the underlying distribution of signals that different review systems draw from differ. This means that anything can happen. In order to gain structure, I restrict attention to reporting rates p that are separable between the impact of the review function and the signal realization.

Definition 4. The propensity to review function $p(r, s)$ is said to be *separable* if $p(r, s) = p_r q(s)$ for some functions $p : \mathbb{N} \cup \{\infty\} \to (0, 1]$ and $q : \mathbb{R} \to [0, 1]$.

If the willingness to review W_i are distributed exponentially, then the resulting propensity to review is separable (following a normalization so that $q(s) \leq 1$).[18] To understand the new forces that emerge when reporting rates are signal-dependent, I now develop an analogue of Theorem 2 in the case that propensities to review are separable. Before I present the result, additional notation is needed. Fix the distribution of taste heterogeneity f and the signal reporting rate q. Define the functional E as the integral of a function g with respect to the measure whose density is given by $q \cdot f$:

$$E(g) := \int_{-\infty}^{\infty} g(s)q(s)f(s)ds.$$

Note that E is not an expectation because $q \cdot f$ is not probability distribution (unless $q(s) = 1$ almost everywhere). Similarly, given a review system r, for each review $\tilde{r} \in \mathcal{R}_r$, define

$$E(g; \tilde{r}) := \int_{-\infty}^{\infty} g(s)\mathbb{1}\{r(s) = \tilde{r}\}q(s)f(s)ds$$

[18] Importantly, q depends on the reviewer's signal and *not* their taste shock ϵ_i. This latter case reduces to the framework of Section 3

the integral with respect to the restricted measure. With this notation, Theorem 2 generalizes to the following result.

Theorem 3. *Assume that f is log-concave and satisfies Assumption 1 and $p(r,s) = p_r q(s)$. Let $\ell_f := \frac{f'}{f}$ denote the linear score of f. The generalization of relative learning for a finite review system r is*

$$\frac{p_{r_f}}{p_r} \cdot \lim_{\sigma\to\infty} \frac{\rho(\sigma; r, p_r \cdot q)}{\rho(\sigma; r_f, p_{r_f} \cdot q)} = \frac{\sum_{\tilde{r}\in\mathcal{R}_r} \frac{E(\ell_f;\tilde{r})^2}{E(1;\tilde{r})} + \frac{p_r E(\ell_f)^2}{1-p_r E(1)}}{E\left(\ell_f^2\right) + \frac{p_{r_f} E(\ell_f)^2}{1-p_{r_f} E(1)}}. \tag{4}$$

The optimal k component review is a k-threshold system whose thresholds maximize (4).

The left-hand side of (4) is the generalization of $\kappa(\infty, r)$. In the case that $q \equiv 1$, Theorem 3 generalizes Lemma 2 to non-threshold finite reviews. The sum in the numerator and the first term of the denominator are the direct generalizations of the numerator and denominator from Lemma 2.[19]

The impact of null reviews is the last term of the numerator and denominator. Importantly, the information contained in null reviews depends on the rate at which a review is submitted. To illustrate, consider Fig. 6. The top two panels show the distribution of submitted reviews. As p_r increases, these distributions are scaled. However, the distribution of null reviews (shown in the bottom panels) do not simply scale. The mass of null reviews becomes more similar in the two states as fewer reviewers report ($p_r \to 0$) since $1 - q(s)p_r$ approaches 1 uniformly (drowning out asymmetry due to q). Hence, null reviews become less informative as p_r decreases. In the case that $p_r \approx 0$, null reviews are uninformative. In this setting, signal-independent reporting rates approximate the more general model apart from a change in measure.

[19] It is impossible to bound (4) above. It is possible to construct examples where q is large for uninformative reviews, and near zero for very informative reviews. If p_r is large and p_{r_f} is small, the value in (4) can be made arbitrarily large in these examples. As discussed below, as $p_r, p_{r_f} \to 0$, the bound of 1 reappears. The full-detail review system is preferred to any review system r if $p_{r_f} > p_r$.

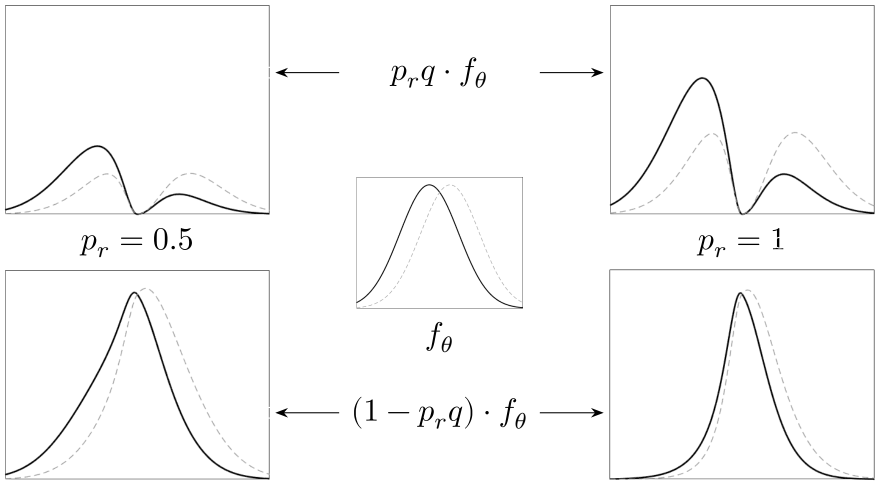


Figure 6: Distributions of reviews for $f_\theta(s)$ normal, with $\frac{\mu}{\sigma} = 0.25$, and $q(s) = 1 - \frac{1}{1+2(x^2+4\mathbb{1}\{x\leq 0\}x^2)}$ and for $p_r = 0.5$ (left), 1 (right). The top panels show the distribution of submitted reviews, and the bottom panels show the distribution of null reviews.

5.2 Comparing Numbers of Reviews versus Probability of Reviewing

An alternative to comparing rates of reporting is to directly compare n_r reviews from system r with $n_{r'}$ reviews from system r'.[20] This comparison falls to consumers who must decide between multiple types of reviews on a website: should they read written reviews, or instead just consider the 5-star rating? The consumer's preferences and the platform's preferences over review systems are related, they are not identical.

Consider two different review systems, r and r', with $\nu(r) > \nu(r')$ and $p_r < p_{r'}$. The platform must choose a review system before information is collected. This means that it must consider the possibility of not receiving any (or receiving very few) reviews. This happens with higher probability under review system r. However, on average, there are many reviews of type r submitted. For a consumer deciding between Np_r reviews from system r and $Np_{r'}$ reviews from system r', the risk of seeing few reviews is not a concern. This means that varying p_r has a larger effect on the platform's preferences over review systems than it does on the consumer's (on average). When reviewers are informationally small, this difference disappears and the two orders agree. Appendix B formalizes the model and the differences, and

[20]This is the traditional approach taken in the literature (Chernoff 1952; Moscarini and Smith 2002; Mu et al. 2021).

shows these relationships.

5.3 Multi-Dimensional Information

In practice, even simple products like toasters have multiple dimensions of quality. While my results do not directly speak to this setting, they do suggest a naive approach. It is always possible for the platform to isolate different dimensions of product quality, and ask a simple question along each dimension. My results suggest that, because of the performance of binary systems in the one-dimensional case, even this naive approach performs well. An interesting question for future work is when it is optimal to isolate dimensions in this way, and when it is better to combine information across dimensions.

5.4 More than Two States

When there are more than two states of the world, my results can be extended in the standard way. First, partition states by the optimal action in that state. Then, the rate of learning for the platform is determined by the two states across elements of the partition that are hardest to distinguish. In the case of binary action environments (fire/keep on, recommend/do not recommend, or when flagging a problem), the platform's problem becomes one of differentiating two states, mirroring my specification. An important difference is that, here, the optimal review system depends on the platform's decision problem, because its problem determines which two states are must be distinguished. In settings with a richer action space, more care must be taken. It is possible that a review system that distinguishes two elements of the partition well might poorly distinguishes a third. In this case, more complicated review systems may be preferred.

6 Conclusion

My results show how a principal optimally designs review systems, abstracting from the systems' implementation. In practice, the platform must also consider how to encourage reviewers to leave reviews as suggested. Common review systems, like 5-star reviews, are likely difficult to fine-tune because of reviewers' extensive experience with similar systems: reviewers are likely to default to their own, private, thresholds.[21] Binary systems, because of their simplicity, are easier to carefully design. In practice,

[21] This is a major mechanism driving the results of Botelho et al. (2025).

many different binary reviews are used—“did you like the product?”, “would you buy the product again?”, “would you recommend the product to a friend/colleague?”—all of which induce different thresholds. If implementation of complex review systems is difficult, then there is an additional benefit to using simple review systems.

Whenever a principal elicits information from agents, it must consider the trade-off between the quality of information that it elicits and the burden it places on agents. If the principal asks for too much information, agents may choose to opt-out and not provide any information. My model characterizes this trade-off. I show that learning is sensitive to the design of review systems and that the principal must carefully consider reviewers' information. When the platform optimally elicits information even simple review systems perform well. However, the optimal review system depends on reviewers' information. If reviewers are sufficiently heterogeneous, the symmetric “good” or “bad” review system performs well. On the other hand, if reviewers' tastes are homogeneous, this naive binary review system performs poorly. Instead, the optimal binary review system is asymmetric: the platform isolates informative signals in one review at the cost of the other review.

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A Proofs

All proofs for results in the main text are included in this appendix.

A.1 Proofs for Section 3

In this appendix I provide the proof of the major results from Section 3. Before I prove the main result, I first state the following result from Torgersen (1981):

Theorem (Torgersen 1981). *Suppose that reviews are always submitted (i.e., $p_r = 1$ for all review systems r). Fix two review systems r, r' with $\nu(r) > \nu(r')$ and a decision problem for the platform. There exists an $\overline{N}$ such that for all $N \geq \overline{N}$, such that $u^*(r, N) > u^*(r', N)$.*

Proof of Theorem 1. For ease of notation, write $\nu_r := \nu(r)$ to denote the learning efficiency of a review r. The pair (p_r, r) can be thought of as defining a statistical experiment over the two states of the world.

In order to apply the above theorem to this setting, we must translate (p_r, r) into a statistical experiment without stochastic reporting, and show that its learning efficiency is given by $p_r \nu_r$, where ν_r is the learning efficiency of the statistical experiment $(1, r)$.

Let $\gamma_L^{(p_r,r)}$ be the measure defined over $\mathcal{R}_r \cup \{\emptyset\}$ representing the distribution of reviews in state L. That is,

$$\begin{aligned}\gamma_L^{(p_r,r)}(B) :=& p_r \cdot \int_{\{s:r(s)\in B\}} f_L(s)ds + (1-p_r)\cdot \mathbb{1}\{\emptyset \in B\}\\ =& p_r \cdot \gamma_L^r(B \cap \mathcal{R}_r) + (1-p_r)\cdot \mathbb{1}\{\emptyset \in B\}.\end{aligned}$$

The only difference between this measure and the measure γ_L^r is the weight $1-p_r$ placed on $\{\emptyset\}$, which represents a null review (a reviewer choosing to not submit a review). Similarly,

$$\gamma_H^{(p_r,r)}(B) := p_r \cdot \gamma_H^r(B \cap \mathcal{R}_r) + (1-p_r)\cdot \mathbb{1}\{\emptyset \in B\}.$$

For a given value of λ, it is the case that

$$\begin{aligned}\int_{\mathcal{R}_r\cup\{\emptyset\}} \left(d\gamma_L^{(p_r,r)}\right)^{\lambda}\left(d\gamma_H^{(p_r,r)}\right)^{1-\lambda} =& \int_{\mathcal{R}_r} \left(p_r \cdot d\gamma_L^r\right)^{\lambda}\left(p_r \cdot d\gamma_H^r\right)^{1-\lambda} + (1-p_r)\\ =& p_r \int_{\mathcal{R}_r} \left(d\gamma_L^r\right)^{\lambda}\left(d\gamma_H^r\right)^{1-\lambda} + 1 - p_r.\end{aligned}$$

Hence,

$$\min_{\lambda\in[0,1]} \int_{\mathcal{R}_r\cup\{\emptyset\}} \left(d\gamma_L^{(p_r,r)}\right)^{\lambda}\left(d\gamma_H^{(p_r,r)}\right)^{1-\lambda} = 1 - p_r\left(1 - \min_{\lambda\in[0,1]}\int_{\mathcal{R}_r} \left(d\gamma_L^r\right)^{\lambda}\left(d\gamma_H^r\right)^{1-\lambda}\right).$$

Concluding,

$$\nu_{(p_r,r)} = 1 - \min_{\lambda\in[0,1]} \int_{\mathcal{R}_r\cup\{\emptyset\}} \left(d\gamma_L^{(p_r,r)}\right)^{\lambda}\left(d\gamma_H^{(p_r,r)}\right)^{1-\lambda} = p_r \cdot \nu_r,$$

proving the claim. □

The proof of Lemma 1 requires additional mechanics. I first prove a similar result in posterior space, and then show that the result applies in signal space when f is log-concave.

Let $\overline{f}(s) = \frac{1}{2}\left(f_L(s) + f_H(s)\right)$ be the ex-ante distribution of signals under a symmetric prior. Moreover, let $\pi(s) := \frac{f_L(s)}{f_L(s)+f_H(s)}$ be the posterior belief placed on state

L after observing signal s when there is a symmetric prior. A straightforward computation shows that the learning efficiency under the full-detail review system is an expectation that depends on the posterior and the ex ante distribution:

$$\nu(r_f) = 1 - \min_{\lambda \in [0,1]} 2 \int \pi(s)^\lambda (1 - \pi(s))^{1-\lambda} \overline{f}(s) ds.$$

Associated with each signal s is a unique posterior π. Let γ_π be the measure on $[0, 1]$ that reflects the distribution of posteriors that is associated with $\overline{f}$.[22] That is,

$$\gamma_\pi(B) := \frac{1}{2} \int_{\{s:\pi(s)\in B\}} f_L(s) + f_H(s) ds.$$

The learning efficiency of the full-detail review system can be written as

$$\nu(r_f) = 1 - \min_{\lambda \in [0,1]} 2 \int \pi^\lambda (1 - \pi)^{1-\lambda} \gamma_\pi(d\pi).$$

This structure is preserved when other review systems are used, as long as the review system treats posteriors consistently. That is, as long as it treats two signals that induce the same posterior the same. To this end, call a review system r *consistent* if, for almost all signals $s, s' \in \mathbb{R}$ such that $\pi(s) = \pi(s')$, $r(s) = r(s')$. Notice that if γ_π has no atoms, then it is without loss to restrict attention to consistent review systems.

The representation of $\nu(r_f)$ in terms of the distribution of posteriors extends to arbitrary consistent review systems. The difference is that instead of the function $\pi^\lambda (1 - \pi)^{1-\lambda}$ being evaluated at each π, the review system r groups all posteriors that are mapped to the same review.

Lemma A.1. Let r be any consistent review system. Then,

$$\nu(r) = 1 - \min_{\lambda \in [0,1]} 2 \int \left(\mathbb{E}_{\gamma_\pi} \left[\tilde{\pi} | r(\tilde{\pi}) = r(\pi) \right] \right)^\lambda \left(1 - \mathbb{E}_{\gamma_\pi} \left[\tilde{\pi} | r(\tilde{\pi}) = r(\pi) \right] \right)^{1-\lambda} \gamma_\pi(d\pi).$$

Proof. I show this in the case that r is a finite review system, and when γ^r has full support on $\mathcal{R}_r$, since the notation is easier in this case. First, observe that

[22] Although γ_π is the measure of posteriors on state L, I use γ_π to indicate that this is the ex-ante measure, and not the measure conditional on state L.

$f_H(s) = f_L(s)\frac{1-\pi(s)}{\pi(s)}$. This means that $\nu(r)$ can be written as as

$$
\begin{aligned}
1-\nu(r) &= \min_{\lambda\in[0,1]} \sum_{\tilde{r}\in\mathcal{R}_r} \mathbb{P}_L(r^{-1}(\tilde{r}))^{\lambda}\mathbb{P}_H(r^{-1}(\tilde{r}))^{1-\lambda} \\
&= \min_{\lambda\in[0,1]} \sum_{\tilde{r}\in\mathcal{R}_r} \left(\int_{\{s\in r^{-1}(\tilde{r})\}} \frac{\pi(s)}{(1-\pi(s))} f_H(s)ds\right)^{\lambda} \cdot \left(\int_{\{s\in r^{-1}(\tilde{r})\}} f_H(s)ds\right)^{1-\lambda}
\end{aligned}
$$

If $\pi(s)$ is injective, then by straightforward computation it is easy to show that $\frac{1}{1-\pi}f_H(s) = 2\gamma_\pi(d\pi(s))$ and $f_H(s) = 2(1-\pi(s))\gamma_\pi(d\pi(s))$. In general, as long as r is consistent, even if $\pi(s)$ is not injective, then for each λ

$$
\begin{aligned}
&\sum_{\tilde{r}\in\mathcal{R}_r} \mathbb{P}_L(r^{-1}(\tilde{r}))^{\lambda}\mathbb{P}_H(r^{-1}(\tilde{r}))^{1-\lambda} \\
&\quad= \sum_{\tilde{r}\in\mathcal{R}_r} 2\left(\int_{\tilde{\pi}\in r^{-1}(\tilde{r})} \tilde{\pi}\gamma_\pi(d\tilde{\pi})\right)^{\lambda} \cdot \left(\int_{\tilde{\pi}\in r^{-1}(\tilde{r})} (1-\tilde{\pi})\gamma_\pi(d\tilde{\pi})\right)^{1-\lambda} \\
&\quad= \sum_{\tilde{r}\in\mathcal{R}_r} 2\frac{\left(\int_{\tilde{\pi}\in r^{-1}(\tilde{r})} \tilde{\pi}\gamma_\pi(d\tilde{\pi})\right)^{\lambda} \cdot \left(\int_{\tilde{\pi}\in r^{-1}(\tilde{r})} (1-\tilde{\pi})\gamma_\pi(d\tilde{\pi})\right)^{1-\lambda}}{\int_{\tilde{\pi}\in r^{-1}(\tilde{r})} \gamma_\pi(d\tilde{\pi})} \int_{\tilde{\pi}\in r^{-1}(\tilde{r})} \gamma_\pi(d\tilde{\pi}) \\
&\quad= \sum_{\tilde{r}\in\mathcal{R}_r} 2\left(\mathbb{E}_{\gamma_\pi}\left[\pi|r(\pi)=\tilde{r}\right]\right)^{\lambda}\left(1-\mathbb{E}_{\gamma_\pi}\left[\pi|r(\pi)=\tilde{r}\right]\right)^{1-\lambda} \int_{\tilde{\pi}\in r^{-1}(\tilde{r})} \gamma_\pi(d\tilde{\pi})
\end{aligned}
$$

From here, the conclusion is reached by minimizing over λ. □

Notice that here the measure with respect to which the integral is taken is simply $\gamma_\pi(d\pi)$, which is *independent* of r. This means that, fixing λ, the loss in information is from the pre-evaluation of $2\pi^{\lambda}(1-\pi)^{1-\lambda}$. Since, for any λ, this function is concave, information loss is exactly due to Jensen's inequality.

If r is a consistent review system with k components, it is a partition of the space $[0,1]$ into k sections. Let $\{\Pi_1, ..., \Pi_k\}$ be the partition of the interval. Then $\nu(r)$ can be rewritten as

$$
\nu(r) = 1 - \min_{\lambda\in[0,1]} 2\sum_{i=1}^{k}\left(\mathbb{E}_{\gamma_\pi}\left[\pi|\pi\in\Pi_i\right]\right)^{\lambda}\left(1-\mathbb{E}_{\gamma_\pi}\left[\pi|\pi\in\Pi_i\right]\right)^{1-\lambda}\gamma_\pi\left(\Pi_i\right).
$$

In order to show that threshold rules are optimal in signal space, I show that they are optimal in threshold space.

Definition A.1. A (posterior) review system r is called a *k-threshold* review system if $r^{-1}(\tilde{r})$ is convex (except perhaps on a set of measure zero) for each $\tilde{r} \in \mathcal{R}_r$.

Optimal review systems are necessarily threshold rules in posterior space.

Lemma A.2. Let r be an optimal (posterior) review system with k reviews. Then r is k-threshold review system.

Proof. I show that if r is a finite review system with k reviews that is not a threshold rule, then there exists a review with k reviews that outperforms it. Moreover, I show that this holds point-wise for each λ, so that certainly it holds over the minimum of λ.

Consider two reviews r^1 and r^2 that cannot be separated by any threshold. Let $\pi_i = \mathbb{E}[\pi | r(\pi) = r^i]$. Without loss assume that $\pi_1 \leq \pi_2$.

Consider any point $\overline{\pi}$ that lies between π_1 and π_2 (if they are equal, then $\overline{\pi} = \pi_1 = \pi_2$). By assumption, $\overline{\pi}$ is not a threshold for these two reviews. In particular, this means that because $\pi_1 \leq \overline{\pi} \leq \pi_2$, there exists a set Π_1 with positive measure such that for all $\pi \in \Pi_1$, $\pi > \overline{\pi}$ and $r(\pi) = r^1$. That is, consider the following two sets

$$\begin{aligned} \Pi_1 &:= \left\{\pi | r(\pi) = r^1 \text{ and } \pi > \overline{\pi}\right\} \quad \text{and} \\ \Pi_2 &:= \left\{\pi | r(\pi) = r^2 \text{ and } \pi < \overline{\pi}\right\}. \end{aligned}$$

Because $\overline{\pi}$ is not a threshold for r^1, r^2, and $\pi_1 \leq \overline{\pi} \leq \pi_2$, these sets both must have positive measure. Let ϵ bound the measures of Π_1, Π_2 below. Select a subset of each that has mass ϵ: Π_1^ϵ and Π_2^ϵ respectively. Now, define a new review function r' as follows:

$$r'(\pi) = \begin{cases} r(\pi) & \text{if} \quad \pi \notin \Pi_1^\epsilon, \Pi_2^\epsilon \\ r^1 & \text{if} \quad \pi \in \Pi_2^\epsilon \\ r^2 & \text{if} \quad \pi \in \Pi_1^\epsilon. \end{cases}$$

That is, r' agrees with r, except that it swaps the Π_i^ϵ's. I show that we necessarily have $\nu(r') \geq \nu(r)$. As mentioned, I show that this holds for each λ. To that end, fix a value of $\lambda \in (0, 1)$, and let $\varphi(\pi) := \pi^\lambda (1 - \pi)^{1-\lambda}$. I now show that the iterated expectation of $\varphi(\pi)$ on the partition induced by r' is lower than on the partition induced by r. That is, I aim to show that

$$\sum_{i=1}^{k} \varphi\left(\mathbb{E}_{\gamma_\pi}\left[\pi | r(\pi) \in r^i\right]\right) \gamma_\pi(r^{-1}(r^i)) \geq \sum_{i=1}^{k} \varphi\left(\mathbb{E}_{\gamma_\pi}\left[\pi | r'(\pi) \in r^i\right]\right) \gamma_\pi((r')^{-1}(r^i)).$$

Notice now that the two agree except on r^1 and r^2, so that the sum simplifies. Let $\pi'_i = \mathbb{E}[\pi | r'(\pi) = r^i]$ denote the new conditional expectations. By construction, $\pi_1 > \pi'_1$ and $\pi_2 < \pi'_2$. Add a linear function $h(\pi)$ to $\varphi(\pi)$ such that the unique maximum of $h + \varphi$ is obtained at $\overline{\pi}$. Note that this is possible because φ is strictly concave. In particular, this means that $h + \varphi$ is strictly increasing for $\pi < \overline{\pi}$ and strictly decreasing for $\pi > \overline{\pi}$. Then, because h is linear, it is the case that

$$\begin{aligned} & \sum_{i=1}^{2} \varphi\left(\pi_i\right) \gamma_\pi(r^{-1}(r^i)) \geq \sum_{i=1}^{2} \varphi\left(\pi'_i\right) \gamma_\pi((r')^{-1}(r^i)) \\ \iff \quad & \sum_{i=1}^{2} \left(\varphi\left(\pi_i\right) + h(\pi_i)\right) \gamma_\pi(r^{-1}(r^i)) \geq \sum_{i=1}^{2} \left(\varphi\left(\pi'_i\right) + h(\pi'_i)\right) \gamma_\pi((r')^{-1}(r^i)). \end{aligned}$$

As observed, it is the case that $\overline{\pi} \geq \pi_1 > \pi'_1$. This means that

$$\varphi\left(\pi_1\right) + h(\pi_1) > \varphi\left(\pi'_1\right) + h(\pi'_1).$$

Similarly,

$$\varphi\left(\pi_2\right) + h(\pi_2) > \varphi\left(\pi'_2\right) + h(\pi'_2).$$

Since by construction it is the case that $\gamma_\pi(r^{-1}(r^i)) = \gamma_\pi((r')^{-1}(r^i))$ for $i = 1, 2$, this means that we have achieved a strict improvement. Hence it must be the case that $\nu(r) < \nu(r')$. □

Proof of Lemma 1. If f is log-concave, then for any level of noise $\frac{f_H}{f_L}$ is monotonically increasing. Consider the optimal posterior review system r_π. The optimal review system in signal space is given by $r_\pi \circ \pi(s)$. Since π is monotonic (because $\frac{f_H}{f_L}$ is increasing), and r_π is a k-threshold system by Lemma A.2, it must be the case that $r_\pi \circ \pi$ is a threshold system. □

Proof of Lemma 2. The proof follows from an application of Theorem 3, to the case that $q \equiv 1$ and the review system r is a threshold system. To see this, observe that (with the notation of the proof of Theorem 3), if $\tilde{r}$ consists of those signals that fall between τ_{i-1} and τ_i, it is the case that

$$H'_+(0,\tilde{r})^2 = \left(\int_{\tau_{i-1}}^{\tau_i} p_r f'(s) ds\right)^2 = p_r^2 \left(f(\tau_i) - f(\tau_{i-1})\right)^2 \quad \text{and}$$
$$H_+(0,\tilde{r}) = \left(\int_{\tau_{i-1}}^{\tau_i} p_r f(s) ds\right) = p_r \left(F(\tau_i) - F(\tau_{i-1})\right).$$

The result follows from simply taking the ratio. □

Proof of Theorem 2. This follows from an immediate application of Lemmas 1 and 2. □

A.2 Proofs for Section 4

In this appendix I provide the proofs of the results in Section 4. To begin the analysis, some preliminary calculations are needed. Important for the calculations below is the Gamma function.

Definition A.2. Denote by $\Gamma(\alpha)$ the standard Gamma function, given by

$$\Gamma(\alpha) := \int_0^\infty x^{\alpha-1} e^{-x} dx.$$

Let $\Gamma(\alpha, t)$ denote the upper incomplete Gamma function:

$$\Gamma(\alpha, t) := \int_t^\infty x^{\alpha-1} e^{-x} dx.$$

Finally, let $\gamma(\alpha, t)$ denote the lower incomplete Gamma function

$$\gamma(\alpha, t) := \int_0^t x^{\alpha-1} e^{-x} dx = \Gamma(\alpha) - \Gamma(\alpha, t).$$

Lemma A.3. It is the case that

$$f_\alpha(x) = \frac{\alpha}{2\Gamma(1/\alpha)} e^{-|x|^\alpha}.$$

Moreover, the Fisher information of f_α is given by

$$I_{f_\alpha}(0) = \frac{\alpha^2}{\Gamma(1/\alpha)}\Gamma\left(2 - \frac{1}{\alpha}\right).$$

Proof. The first part of the statement just ensures that f_α is a probability distribution.

$$\int_{-\infty}^{\infty} e^{-|x|^\alpha} dx = 2\int_0^{\infty} e^{-x^\alpha} dx.$$

Make now the substitution $u = x^\alpha$. Then, $\frac{1}{\alpha}u^{\frac{1}{\alpha}-1}du = dx$, so that

$$2\int_0^{\infty} e^{-x^\alpha} dx = 2\int_0^{\infty} \frac{1}{\alpha}u^{\frac{1}{\alpha}-1}e^{-u}du = \frac{2}{\alpha}\Gamma\left(\frac{1}{\alpha}\right).$$

This proves the first claim. To show the second claim, note that f_α satisfies standard regularity conditions for all $\alpha \geq 1$, so that its Fisher Information is also equal to

$$I_f = \mathbb{E}\left[\left(\frac{f'(s)}{f(s)}\right)^2\right] = -\mathbb{E}\left[\frac{\partial^2}{\partial s^2}\log(f(s))\right].$$

This means that it is the case that

$$\begin{aligned} I_{f_\alpha} =& \frac{\alpha}{2\Gamma(1/\alpha)}\int_{-\infty}^{\infty}\left(\frac{\partial^2}{\partial x^2}|x|^\alpha\right)e^{-|x|^\alpha}dx = \frac{\alpha}{\Gamma(1/\alpha)}\int_0^{\infty}\left(\frac{\partial^2}{\partial x^2}x^\alpha\right)e^{-x^\alpha}dx \\ =& \frac{\alpha}{\Gamma(1/\alpha)}\int_0^{\infty}\alpha(\alpha-1)x^{\alpha-2}e^{-x^\alpha}dx. \end{aligned}$$

Now, perform again the substitution of $u = x^\alpha$.

$$\begin{aligned} =& \frac{\alpha}{\Gamma(1/\alpha)}\int_0^{\infty}\alpha(\alpha-1)\frac{1}{\alpha}u\cdot u^{-\frac{2}{\alpha}}\cdot u^{\frac{1}{\alpha}-1}e^{-u}dx \\ =& \frac{\alpha}{\Gamma(1/\alpha)}\int_0^{\infty}(\alpha-1)u^{-\frac{1}{\alpha}}e^{-u}dx = \frac{\alpha(\alpha-1)}{\Gamma(1/\alpha)}\Gamma\left(1-\frac{1}{\alpha}\right). \end{aligned}$$

Since it is the case that $\Gamma(\alpha+1) = \alpha\Gamma(\alpha)$,

$$\Gamma\left(1-\frac{1}{\alpha}\right) = \frac{\alpha}{\alpha-1}\Gamma\left(2-\frac{1}{\alpha}\right),$$

Concluding,

$$I_{f_\alpha}(0) = \frac{\alpha^2}{\Gamma(1/\alpha)}\Gamma\left(2 - \frac{1}{\alpha}\right).$$

$\square$

In all proofs, denote $\kappa_{f_\alpha}(\infty; r_\tau)$ by $\kappa(\alpha;\tau)$. This is just done to condense notation: κ is a function of homogeneity α and the threshold τ of the review system.

Proof of Proposition 1. First, a simple computation shows that $\kappa(\alpha;0) = \frac{1}{\Gamma(\frac{1}{\alpha})\Gamma(2-\frac{1}{\alpha})}$. This shows that $\kappa(1;0) = 1$, as $\Gamma(1) = 1$.

I now proceed to the other claims. Let $\psi(\alpha)$ denote the digamma function

$$\psi(\alpha) := \frac{\Gamma'(\alpha)}{\Gamma(\alpha)}.$$

It is well-known that ψ is strictly increasing on $(0,\infty)$.

Now, let $\left(\frac{1}{\kappa(\alpha;0)}\right)'$ denote $\frac{\partial}{\partial\alpha}\frac{1}{\kappa(\alpha;0)}$. A straightforward computation yields that

$$\left(\frac{1}{\kappa(\alpha;0)}\right)' = \frac{\Gamma\left(\frac{1}{\alpha}\right)\Gamma\left(2-\frac{1}{\alpha}\right)}{\alpha^2}\left(\psi\left(2-\frac{1}{\alpha}\right) - \psi\left(\frac{1}{\alpha}\right)\right).$$

Since ψ is strictly increasing, it is immediately seen that $\kappa'(\alpha;0) < 0$.

It is left to show that $\lim_{\alpha\to\infty}\kappa(\alpha;0) = 0$. To see this, observe that $\Gamma(1) = 1$ and $\lim_{\alpha\to 0}\Gamma(\alpha) = \infty$.

$\square$

Proof of Proposition 2. As noted, I restrict attention to $\tau \in [0,\infty)$ for the proof. The analogous proofs apply for $\tau \le 0$. For ease of reference I restate that the objective that the platform aims to maximize is given by.

$$g_\alpha(\tau) := \frac{f_\alpha(\tau)^2}{F_\alpha(\tau)(1 - F_\alpha(\tau))} \tag{5}$$

I proceed in several steps. I first show that $\tau^*(\alpha) \neq 0$ for $\alpha > 2$, and that it 0 is a candidate for $\tau^*(\alpha)$ for $\alpha \le 2$.

Lemma A.4. For $\alpha \leq 2$, 0 is a local maximum of $g_\alpha(\tau)$. For $\alpha > 2$, 0 is a local minimum of $g_\alpha(\tau)$.

Proof. The derivative of (5) with respect to τ is given by (dropping the subscript α)

$$\frac{f(\tau)}{F(\tau)(1-F(\tau))}\left[2f'(\tau) - f(\tau)^2\left(\frac{1}{F(\tau)} - \frac{1}{1-F(\tau)}\right)\right].$$

As f is of full support, $\frac{f(\tau)}{F(\tau)(1-F(\tau))} > 0$, and so can be ignored when searching for a critical value, and hence a maximum. The first order condition to be a local maximum then becomes

$$2f'(\tau) = f(\tau)^2\left(\frac{1}{F(\tau)} - \frac{1}{1-F(\tau)}\right). \tag{6}$$

For any $\alpha > 1$, 0 is a solution to (6), and hence a candidate for a maximizer. This is as $f'_\alpha(0) = 0$ and $F_\alpha(0) = \frac{1}{2}$.

This means that to verify the claim the second order condition is needed. At 0, it can be verified that the second-order condition for a maximizer (since $F_\alpha(0) = \frac{1}{2}$) is

$$\frac{1}{2}f_\alpha(0)^4 + \frac{1}{8}f'_\alpha(0)^2 + \frac{1}{8}f_\alpha(0)f''_\alpha(0) \leq 0.$$

Notice, for all $\alpha > 2$, we have that $f'_\alpha(0) = 0$ and $f''_\alpha(0) = 0$. This means that, as $f_\alpha(0) > 0$, for all $\alpha > 2$, $\tau = 0$ is a local minimum, and hence is not optimal. Similarly, for $\alpha < 2$, it is the case that $\lim_{\tau\to 0} f''_\alpha(\tau) = -\infty$, while $f'_\alpha(0), f_\alpha(0) > 0$ This means that 0 is indeed a local maximum for $\alpha < 2$. The case $\alpha = 2$ can be computed numerically, and it can be seen that 0 is a local maximum in this case. □

I now show that there is at most one possible candidate for the maximizer of $g_\alpha(\tau)$ on $\tau \in [0, \infty)$. This shows the uniqueness of $\tau^*(\alpha)$.

Lemma A.5. The following are true:

(i) For $\alpha > 2$, there is exactly one value of $\tau \in (0, \infty)$ that satisfies (6) and it is a local maximum of $g_\alpha(\tau)$; and

(ii) For $\alpha \leq 2$, no value of $\tau \in (0, \infty)$ satisfies (6).

Proof. Throughout I drop the subscript of α for notational clarity.

Case 1: $\alpha > 2$.
Straightforward algebra allows (6) to be rewritten as

$$\frac{2f'(\tau)}{1-2F(\tau)} = \frac{f^2(\tau)}{F(\tau)(1-F(\tau))}. \tag{7}$$

Notice that the right-hand side of (7) is exactly $g(\tau)$. This means that the value of g can also be used to determine the sign of its derivative. Specifically, the critical points of the objective are exactly those that intersect the curve

$$w(\tau) := \frac{2f'(\tau)}{1-2F(\tau)}.$$

That is, $g'(\tau) = 0$ if and only if $g(\tau) = w(\tau)$. Moreover,

$$g'(\tau) > 0 \iff g(\tau) > w(\tau).$$

Due to this relationship, the properties of $w(\tau)$ imply properties of $g(\tau)$. In particular, I now show that $w(\tau)$ has one unique maximum on $(0, \infty)$. To do this, I iterate on the process above. The sign of $w'(\tau)$ is determined by the sign of

$$2f''(\tau)(1-2F(\tau)) + 4f'(\tau)f(\tau) \tag{8}$$

in particular, $w'(\tau) = 0$ if and only if

$$w(\tau) = \frac{2f'(\tau)}{1-2F(\tau)} = -\frac{f''(\tau)}{f(\tau)} =: h(\tau).$$

Moreover, $w'(\tau) > 0$ if and only if $h(\tau) > w(\tau)$ (notice that this is the reverse relationship of w and g).

While w and g are difficult to manage because the presence of F in their definition (and there is no closed-form expression for the incomplete Gamma functions), $h(\tau) = (\alpha^2 - \alpha)x^{\alpha-2} - \alpha^2 x^{2\alpha-2}$. This (relative to g and w) is an extremely simple function. It can be seen that for all $\alpha > 2$, h has the following properties: (i) $h(0) = 0$, and $h(\tau) > 0$ for τ in a neighbourhood of 0; (ii) h is single-peaked; (iii) $h(1) < 0$; and (iv) $h(\tau) > w(\tau)$ in a neighbourhood of 0. Properties (i)-(iii) are easily to verify, and property (iv) follows from $w'(\tau) > 0$ in a neighbourhood of 0 (which directly implies that $w(\tau) < h(\tau)$).

Claim 1a: $w(\tau)$ has at one critical value on $(0, \infty)$. It is a local maximum, and this critical value is on $(0, 1)$.

Recall that if $w(\tau) < h(\tau)$, then $w'(\tau) > 0$, and if $w(\tau) > h(\tau)$, then $w'(\tau) < 0$. First, observe that all critical values of w must be on $(0, 1)$ because h is negative of $[1, \infty)$ (so $w(\tau) \neq h(\tau)$ for any $\tau \geq 1$).

Second, let τ_h be the value of τ that maximizes h. Notice that h is increasing on $(0, \tau_h)$ and decreasing on (τ_h, ∞). Suppose now that $w(\tau') = h(\tau')$ for some $\tau' \in (0, \tau_h)$. There must be a smallest value of τ' for which this holds, as $w(\tau)$ is strictly increasing in a neighbourhood of 0. For this τ', it is the case that $w(\tau') = 0$ and $h(\tau') > 0$. This means that for some $\epsilon > 0$, for all $\tau \in (\tau' - \epsilon, \tau')$ it is the case that $w(\tau) > h(\tau)$. But τ' is the smallest positive value where the two cross, which is a contradiction. This means that it cannot be the case that $w'(\tau) = 0$ when $h'(\tau) > 0$. This is as, for w to cross h when h is increasing it must be the case that w is larger than h before the crossing.

Since eventually $w(\tau) > h(\tau)$, it must be the case then that either $w = h$ at the peak of h, or on the region where h is decreasing. In either case it must be the case that w goes from increasing to decreasing, and can not cross h again. This proves the claim.

Claim 1b: $g(\tau)$ has a most one critical value on $(0, \infty)$. It is a local maximum, and this critical value is on $(0, 1)$.

Recall that, if $g(\tau) > w(\tau)$, then $g'(\tau) > 0$, and if $g(\tau) < w(\tau)$, then $g'(\tau) < 0$. Notice that because zero is a local minimum of $g(\tau)$ (by Lemma A.4), $g(\tau)$ is increasing in a neighbourhood of zero (since we are concerned only with $\tau \geq 0$). Moreover, it is the case that $g(0) > 0 = w(0)$. This second equality holds because $f'(0) = 0$. Denote by τ_w the value of τ that maximizes w on $(0, \infty)$.

Now, suppose that $g(\tau') = w(\tau')$ for some $\tau' \in (\tau_w, \infty)$. This implies that $g(\tau) > w(\tau)$ for $\tau = (\tau', \tau' + \epsilon)$ and some $\epsilon > 0$ (since w is decreasing and g has derivative zero). However, this means that $g'(\tau) > 0$ on this region. Because w is decreasing on this region, it implies that $g'(\tau) > 0$ on (τ', ∞), contradicting that $\lim_{\tau \to \infty} g(\tau) = 0$. This means that the two cannot intersect when $w'(\tau) < 0$.

By a similar logic, it must be the case that $g(\tau) = w(\tau)$ for some $\tau \in (0, \tau_w]$. Call the first point where they intersect τ^* (well-defined because they are not equal at 0). Notice that for all $\tau \in (\tau^*, \tau^* + \epsilon)$, it must be the case that g is decreasing.[23] But, g must continue to decrease until they intersect again. So, they cannot intersect until w decreases. We have, however, ruled out intersection in that case, so it must be that τ^* is unique, proving the claim and finishing Case 1.

Case 2: $\alpha < 2$.
In the case of $\alpha < 2$ showing that there is no critical value on $(0, \infty)$ is simpler and so I omit the details.

In this case, (i) $\lim_{\tau \to 0} w(\tau) = \infty$ and (ii) $h(\tau)$ is monotonically decreasing. These jointly imply that $w(\tau)$ is monotonically decreasing to 0. This in turn implies that $g(\tau)$ is monotonically decreasing to zero (since they cannot intersect when w decreases in a similar argument to above). This shows that in the case that $\alpha < 2$ it must be that $\tau = 0$ is the only possible critical value.

Case 3: $\alpha = 2$.
This is dealt with in a similar manner to Case 2. The major difference is that $w(\tau)$ does not diverge at 0. Instead, it is easy to show that $w(0) = h(0)$ and w is decreasing in a region of 0, so that it must be monotonically decreasing. This case then reduces to Case 2. $\square$

Now that I have showed that τ^* exists and is unique, it is sufficient to look at the solution to the first order condition (6) on $(0, \infty)$. The last things to check are that $\tau^*(\alpha) \to 1$, and the bounding value of κ.

Claim: $\lim_{\alpha \to \infty} \tau^*(\alpha) = 1$.

First, notice that $\tau^*(\alpha) < 1$ for all $\alpha \geq 1$, by Lemma A.5. I show that for all $\tau < 1$, there exists an α_τ such that: $\alpha \geq \alpha_\tau$, $g'_\alpha(\tau) > 0$. To do this, explicitly write the inner term

$$
\begin{aligned}
&2f'(\tau) - f(\tau)^2 \left(\frac{1}{F(\tau)} - \frac{1}{1 - F(\tau)} \right) \\
&\qquad = -\alpha \tau^{\alpha - 1} \cdot \frac{\alpha}{\Gamma(1/\alpha)} e^{-\tau^\alpha}
\end{aligned}
$$

[23] This is obvious in the case that $\tau^* < \tau_w$. In the case that $\tau^* = \tau_w$, it follows from a similar argument to showing that there is no intersection when w is decreasing.

$$+\left(\frac{\alpha}{2\Gamma(1/\alpha)}e^{-\tau^\alpha}\right)^2\left[\frac{\Gamma(1/\alpha)}{\frac{1}{2}\Gamma(1/\alpha,\tau^\alpha)}-\frac{\Gamma(1/\alpha)}{\frac{1}{2}\Gamma(1/\alpha)+\frac{1}{2}\gamma(1/\alpha,\tau^\alpha)}\right].$$

Since onnly the sign of this expression is relevant, it is possible to simplify. It is equivalent to look at the sign of

$$-2\tau^{\alpha-1}+e^{-\tau^\alpha}\left[\frac{1}{\Gamma(1/\alpha,\tau^\alpha)}-\frac{1}{\Gamma(1/\alpha)+\gamma(1/\alpha,\tau^\alpha)}\right]. \tag{9}$$

That is, $g'_\alpha(\tau)>0$ exactly when (9) is positive. When $\tau<1$, the first term converges to 0, so we need only ensure that the second term stays bounded from 0. First, for fixed $\tau<1$, $e^{-\tau^\alpha}\to 1$. Second, as $\alpha\to\infty$, it is easy to verify that $f_\alpha(\tau)\to\frac{1}{2}\mathbb{1}\{\tau\in[-1,1]\}$ (except for the measure zero set of $\{-1,1\}$). This means that

$$\lim_{\alpha\to\infty}\left(\frac{1}{F(\tau)}-\frac{1}{1-F(\tau)}\right)=\left(\frac{2}{\tau+1}-\frac{2}{1-\tau}\right)>0$$

for all fixed $\tau\in(0,1)$. This means that neither portions of the second term go to zero, so that for any τ in this interval, $g'_\alpha(\tau)$ is eventually positive.

This means that $\tau^*(\alpha)$ must eventually grow to 1. Similar arguments show that $\tau^*(\alpha)^\alpha$ converges to the solution of $\Gamma(0,x)-\frac{1}{2x}e^{-x}=0$.

Claim: $\kappa(\alpha;\tau^*(\alpha))<2e^2\int_1^\infty\frac{e^{-x}}{x}dx$.

For all $\alpha\geq 4$, it can be seen that $\kappa(\alpha,1)$ is less than this limit. For $\alpha<4$, using the naive threshold of 0 also returns a value that is less than this limit. This shows the final claim, and completes the proof of Proposition 2. □

Proof of Proposition 3. In order to prove Proposition 3(i), I prove the following, stronger, lemma.

Lemma A.6. $\tau^*(\alpha)^\alpha$ is strictly increasing in α for $\alpha\geq 2$.

Proof. First, notice that

$$\frac{\partial\tau^*(\alpha)^\alpha}{\partial\alpha}>0\iff\left(\frac{\partial^2}{\partial\tau^\alpha\partial\alpha}\frac{f_\alpha^2(\tau)}{F_\alpha(\tau)(1-F_\alpha(\tau))}\right)\Big|_{\tau=\tau^*(\alpha)}>0.$$

This is from the Implicit Function Theorem and $\tau^*(\alpha)$ being a maximizer of the function (so that the second derivative of the function with respect to τ^* is negative).

That is, letting $x = \tau^\alpha$, the sign of

$$\frac{\partial}{\partial \alpha}\left[-2\frac{x}{x^{1/\alpha}}e^x + \left[\frac{1}{\Gamma(1/\alpha, x)} - \frac{1}{\Gamma(1/\alpha) + \gamma(1/\alpha, x)}\right]\right] \tag{10}$$

must be determined at the optimal value of $x^*(\alpha) = \tau^*(\alpha)^\alpha$. Notice that (10) is not exactly equal to the derivative, it is sufficent to look at this term (since the other term will be equal to zero). I now show that (10) is positive at $x^*(\alpha)$.

Another simplification is to replace α with $1/\alpha$. That is, let $\beta := 1/\alpha$. As $\frac{\partial}{\partial \alpha}\beta < 0$, the sign of (10) is the negative the sign of

$$\begin{aligned} b_\beta(x) :=& 2\log(x)\frac{x}{x^\beta}e^x \\ &- \frac{\int_x^\infty y^{\beta-1}\log(y)e^{-y}dy}{\Gamma(\beta, x)^2} + \frac{\int_0^\infty y^{\beta-1}\log(y)e^{-y}dy + \int_0^x y^{\beta-1}\log(y)e^{-y}dy}{(\Gamma(\beta) + \gamma(\beta, x))^2}. \end{aligned}$$

I show that $b_\beta(x^*(\beta)) < 0$ (I abuse notation here by writing $x^*(\beta)$) for $\beta < \frac{1}{2}$. As a final choice to make notation easier, let λ_β denote the measure that is induced by $y^{\beta-1}e^{-y}$ (that is, $d\lambda_\beta(y) = y^{\beta-1}e^{-y}dy$). Then, $b_\beta(x)$ can be written as

$$2\log(x)\frac{x}{x^\beta}e^x - \frac{\int_x^\infty \log(y)d\lambda_\beta(y)}{\Gamma(\beta, x)^2} + \frac{\int_0^\infty \log(y)d\lambda_\beta(y) + \int_0^x \log(y)d\lambda_\beta(y)}{(\Gamma(\beta) + \gamma(\beta, x))^2}. \tag{11}$$

In the notation above, we have that

$$\Gamma(\beta) = \int_0^\infty d\lambda_\beta(y), \quad \Gamma(\beta, x) = \int_x^\infty d\lambda_\beta(y), \quad \text{and} \quad \gamma(\beta, x) = \int_0^x d\lambda_\beta(y).$$

Now, at $x^*(\beta)$, from (10), it is the case that

$$2x^{1-\beta}e^x = \frac{1}{\Gamma(\beta, x)} - \frac{1}{\Gamma(\beta) + \gamma(\beta, x)}.$$

This means that it is possible to write (11) at $x^*(\beta)$ (I write x instead of $x^*(\beta)$ for conciseness) as

$$\log(x)\left(\frac{1}{\Gamma(\beta, x)} - \frac{1}{\Gamma(\beta) + \gamma(\beta, x)}\right)$$

$$
\begin{aligned}
&\quad - \frac{\int_x^\infty \log(y) d\lambda_\beta(y)}{\Gamma(\beta,x)^2} + \frac{\int_0^\infty \log(y) d\lambda_\beta(y) + \int_0^x \log(y) d\lambda_\beta(y)}{(\Gamma(\beta)+\gamma(\beta,x))^2}. \\
&= \frac{\log(x)\int_x^\infty d\lambda_\beta(y)}{\Gamma(\beta,x)^2} - \frac{\log(x)\left(\int_0^\infty d\lambda_\beta(y) + \int_0^x d\lambda_\beta(y)\right)}{\Gamma(\beta)+\gamma(\beta,x)} \\
&\quad - \frac{\int_x^\infty \log(y) d\lambda_\beta(y)}{\Gamma(\beta,x)^2} + \frac{\int_0^\infty \log(y) d\lambda_\beta(y) + \int_0^x \log(y) d\lambda_\beta(y)}{(\Gamma(\beta)+\gamma(\beta,x))^2} \\
&= \frac{\int_x^\infty (\log(x)-\log(y))\, d\lambda_\beta(y)}{\Gamma(\beta,x)^2} \\
&\quad + \frac{\int_0^\infty (\log(y)-\log(x))\, d\lambda_\beta(y) + \int_0^x (\log(y)-\log(x))\, d\lambda_\beta(y)}{(\Gamma(\beta)+\gamma(\beta,x))^2}.
\end{aligned}
$$

Since $\log(\cdot)$ is increasing, it must be that $\int_x^\infty (\log(x)-\log(y))\, d\lambda_\beta(y) < 0$ whenever $\beta < \frac{1}{2}$, so that $x^*(\beta) > 0$. Moreover, it must always be the case that $\Gamma(\beta,x) < \Gamma(\beta)+\gamma(\beta,x)$ whenever $x > 0$. This means that at $x^*(\beta)$ it is the case that

$$
\frac{\int_x^\infty (\log(x)-\log(y))\, d\lambda_\beta(y)}{\Gamma(\beta,x)^2} < \frac{\int_x^\infty (\log(x)-\log(y))\, d\lambda_\beta(y)}{\left(\Gamma(\beta)+\gamma(\beta,x)\right)^2}.
$$

This means that (11) evaluated at $x^*(\beta)$ is less than

$$
\begin{aligned}
&\frac{\int_x^\infty (\log(x)-\log(y))\, d\lambda_\beta(y)}{(\Gamma(\beta)+\gamma(\beta,x))^2} \\
&\quad + \frac{\int_0^\infty (\log(y)-\log(x))\, d\lambda_\beta(y) + \int_0^x (\log(y)-\log(x))\, d\lambda_\beta(y)}{(\Gamma(\beta)+\gamma(\beta,x))^2} \\
&= \frac{2\int_0^x (\log(y)-\log(x))\, d\lambda_\beta(y)}{(\Gamma(\beta)+\gamma(\beta,x))^2} < 0.
\end{aligned}
$$

This proves that $x^*(\beta)$ is strictly decreasing in β, so that $\tau^*(\alpha)^\alpha$ is increasing in α for all $\beta < \frac{1}{2}$. This in turn shows that $\tau^*(\alpha)^\alpha$ is increasing in α for all $\alpha \geq 2$. $\square$

As noted, Proposition 3(i) follows as an immediate corollary of this lemma, since $\tau^*(\alpha) < 1$ for all α.

Proof of Proposition 3(ii):

In what follows, let $P(\alpha,t) := \gamma(\alpha,t)/\Gamma(\alpha)$ denote the normalized lower incomplete Gamma function.

First, recall that $\tau^*(\alpha)^\alpha$ is strictly increasing in α by Lemma A.6. I show the claim separately on two intervals, based off of the behaviour of $f_\alpha(0)$ as a function of α. In what follows, let α^* be defined as the unique value for which $\psi(1/\alpha) + \alpha = 0$. For $\alpha < \alpha^*$, it is the case that $\Gamma(1/\alpha)/\alpha$ is decreasing in α (so that $f_\alpha(0)$ is increasing), and for $\alpha > \alpha^*$ it is the case the $\Gamma(1/\alpha)/\alpha$ is increasing in α (so that $f_\alpha(0)$ is decreasing).

Case 1: $\alpha < \alpha^*$.

First, on this region, for a fixed τ with $0 < \tau < 1$, it must be the case that $f_\alpha(\tau)$ is increasing in α. This is because τ^α is decreasing in α, while $\frac{\alpha}{\Gamma(\alpha)}$ is increasing. Now, $F_\alpha(0) \equiv \frac{1}{2}$, and for all $0 < \tau < \tau'$, $f_\alpha(\tau)$ is increasing in α. This means that, for all $\alpha, \alpha' \in (2, \alpha^*)$ with $\alpha < \alpha'$, it is the case that

$$F_\alpha(\tau^*(\alpha)) < F_{\alpha'}(\tau^*(\alpha)).$$

From Lemma A.6 it is the case that $\tau^*(\alpha') > \tau^*(\alpha)$, so that

$$F_{\alpha'}(\tau^*(\alpha)) < F_{\alpha'}(\tau^*(\alpha')).$$

This shows that $F_\alpha(\tau^*(\alpha))$ is increasing in α for $\alpha < \alpha^*$.

Case 2: $\alpha > \alpha^*$.

Since $\alpha > \alpha^*$ and we have showed that $\tau^*(\alpha)$ is increasing in α, it must be the case that $f_\alpha(\tau^*(\alpha))$ is decreasing in α for $\alpha > \alpha^*$, as by definition $\Gamma(1/\alpha)/\alpha$ is increasing for all $\alpha > \alpha^*$. This means that if

$$\frac{f_\alpha(\tau^*(\alpha))^2}{F_\alpha(\tau^*(\alpha))(1 - F_\alpha(\tau^*(\alpha)))} \tag{12}$$

is increasing in α, then the claim holds in this case. This is because, if (12) is increasing, then either the numerator is increasing or the denominator is decreasing. Since it is not the former, it must be the latter. This implies that $F_\alpha(\tau^*(\alpha))$ is increasing in α, which is what we set out to show. A stronger condition is that

$$\frac{\left(\frac{\alpha}{\Gamma(1/\alpha)}\right)^2}{(1 + P(1/\alpha, \tau))(1 - P(1/\alpha, \tau))} \tag{13}$$

is increasing in α for fixed τ, where $P(1/\alpha,\tau) := \frac{\Gamma(1/\alpha,\tau)}{\Gamma(1/\alpha)}$ is the normalized incomplete Gamma function. This is a stronger condition because it is saying is that if we look at values of τ for each α that make $e^{-\tau^\alpha}$ equal, then these values of τ cause this ratio to increase. In particular, consider evaluating (12) at $\tau^*(\alpha)$ for some α. If (13) is increasing in α, then for all $\alpha' > \alpha$ it must be the case that

$$\frac{f_{\alpha'}(\tau^*(\alpha)^{\alpha/\alpha'})^2}{F_{\alpha'}(\tau^*(\alpha)^{\alpha/\alpha'})(1-F_{\alpha'}(\tau^*(\alpha)^{\alpha/\alpha'}))} > \frac{f_\alpha(\tau^*(\alpha))^2}{F_\alpha(\tau^*(\alpha))(1-F_\alpha(\tau^*(\alpha)))}.$$

Since by optimality it is the case that

$$\frac{f_{\alpha'}(\tau^*(\alpha'))^2}{F_{\alpha'}(\tau^*(\alpha'))(1-F_{\alpha'}(\tau^*(\alpha')))} \geq \frac{f_{\alpha'}(\tau^*(\alpha)^{\alpha/\alpha'})^2}{F_{\alpha'}(\tau^*(\alpha)^{\alpha/\alpha'})(1-F_{\alpha'}(\tau^*(\alpha)^{\alpha/\alpha'}))},$$

this will prove the claim.

I now show that (13) is indeed increasing in α. Perform the switch to $\beta := 1/\alpha$, as the in proof of Lemma A.6. Then, the above can be rewritten as

$$\begin{aligned}&\frac{\left(\frac{1}{\beta\Gamma(\beta)}\right)^2}{(1+P(\beta,x))(1-P(\beta,x))}\\ &= \frac{1}{2\beta^2\Gamma(\beta)}\left(\frac{1}{\Gamma(\beta)+\gamma(\beta,x)}+\frac{1}{\Gamma(\beta)-\gamma(\beta,x)}\right).\end{aligned}$$

Multiplying by 2 and then taking the derivative with respect to β here yields

$$\begin{aligned}&-\frac{2\beta\Gamma(\beta)+\beta^2\int_0^\infty \log(y)d\lambda_\beta(y)}{(\beta^2\Gamma(\beta))^2}\left(\frac{1}{\Gamma(\beta)+\gamma(\beta,x)}+\frac{1}{\Gamma(\beta)-\gamma(\beta,x)}\right)\\ &-\frac{1}{\beta^2\Gamma(\beta)}\left(\frac{\int_0^\infty \log(y)d\lambda_\beta(y)+\int_0^x \log(y)d\lambda_\beta(y)}{(\Gamma(\beta)+\gamma(\beta,x))^2}+\frac{\int_x^\infty \log(y)d\lambda_\beta(y)}{(\Gamma(\beta)-\gamma(\beta,x))^2}\right).\end{aligned}$$

Here I use the definition of $\lambda_\beta(y)$ from the proof of Lemma A.6. The goal is to show that this object is negative. Again let $\Gamma'(\beta)/\Gamma(\beta) =: \psi(\beta)$ and observe that for $\alpha > \alpha^*$ by definition it the case that $\frac{1}{\beta}+\psi(\beta) < 0$. Now, straightforward arithmetic shows that the sign of this is equivalent to the sign of

$$-\frac{\int_0^\infty \left(\log(y)+\frac{2}{\beta}+\psi(\beta)\right)d\lambda_\beta(y)+\int_0^x\left(\log(y)+\frac{2}{\beta}+\psi(\beta)\right)d\lambda_\beta(y)}{(\Gamma(\beta)+\gamma(\beta,x))^2}$$

$$-\frac{\int_x^\infty \left(\log(y)+\frac{2}{\beta}+\psi(\beta)\right) d\lambda_\beta(y)}{(\Gamma(\beta)-\gamma(\beta,x))^2}.$$

I first show that for all $x \in (0,\infty)$ and β, it is the case that

$$-\int_x^\infty \left(\log(y)+\frac{2}{\beta}+\psi(\beta)\right) d\lambda_\beta(y) < 0.$$

Now, consider that on this region of β, by definition it is the case that

$$-\left(\frac{2}{\beta}+\psi(\beta)\right) = 2\left(\frac{1}{\beta}+\psi(\beta)\right)+\psi(\beta) < \psi(\beta).$$

This means that

$$-\int_x^\infty \left(\log(y)+\frac{2}{\beta}+\psi(\beta)\right) d\lambda_\beta(y) < \int_x^\infty \left(-\log(y)+\psi(\beta)\right) d\lambda_\beta(y).$$

Now, the sign of the this right-hand side is the same as the sign of

$$\frac{\int_x^\infty \left(-\log(y)+\psi(\beta)\right) d\lambda_\beta(y)}{\int_x^\infty d\lambda_\beta(y)}.$$

This, however, is clearly decreasing in x, because smaller values of x correspond to the average log value being smaller. Since it is clearly equal to zero at $x = 0$, this shows that

$$-\int_x^\infty \left(\log(y)+\frac{2}{\beta}+\psi(\beta)\right) d\lambda_\beta(y) < 0.$$

This in turn implies that

$$-\frac{\int_x^\infty \left(\log(y)+\frac{2}{\beta}+\psi(\beta)\right) d\lambda_\beta(y)}{(\Gamma(\beta)-\gamma(\beta,x))^2} < -\frac{\int_x^\infty \left(\log(y)+\frac{2}{\beta}+\psi(\beta)\right) d\lambda_\beta(y)}{(\Gamma(\beta)+\gamma(\beta,x))^2}.$$

This means that it is sufficient to check the sign of

$$\begin{aligned}
&-\int_0^\infty \left(\log(y)+\frac{2}{\beta}+\psi(\beta)\right) d\lambda_\beta(y) - \int_0^x \left(\log(y)+\frac{2}{\beta}+\psi(\beta)\right) d\lambda_\beta(y) \\
&\quad -\int_x^\infty \left(\log(y)+\frac{2}{\beta}+\psi(\beta)\right) d\lambda_\beta(y)
\end{aligned}$$

and ensure that it is negative. But this expression is equal to

$$-2\int_0^\infty \left(\log(y)+\frac{2}{\beta}+\psi(\beta)\right) d\lambda_\beta(y) = -\Gamma'(\beta)-\left(\frac{2}{\beta}+\psi(\beta)\right)\Gamma(\beta)<0,$$

where this last inequality follows from the observation above:

$$\left(\frac{2}{\beta}+\psi(\beta)\right)\Gamma(\beta) > -\Gamma'(\beta)$$

This completes the proof in this case. □

Proof of Proposition 4. First, notice that

$$f(s) := f_{\alpha_L,\alpha_H}(s) = \frac{1}{\frac{\Gamma(1/\alpha_L)}{\alpha_L}+\frac{\Gamma(1/\alpha_H)}{\alpha_H}}\left(\mathbb{1}\{s\geq 0\}e^{-s^{\alpha_H}}+\mathbb{1}\{s\leq 0\}e^{-(-s)^{\alpha_L}}\right)$$

Recall that, by definition, $\frac{\Gamma(1/\alpha)}{\alpha}$ is increasing in α for all $\alpha \geq \alpha^*$. This means, in particular, that there is a larger mass of positive signals than negative signals in this case.

What must be shown is that

$$\frac{f(\tau)^2}{F(\tau)(1-F(\tau))}$$

is maximized for a negative value of τ. For ease of notation, let $\alpha := \alpha_H$ and $\alpha' := \alpha_L$. Notice that as before, the numerator is strictly increasing in τ for $\tau < 0$, and strictly decreasing in τ for $\tau > 0$. Since $\alpha' > \alpha$, for small values of τ there is larger persistence in the value of f on the negative side of the distribution.[24] Explicitly, given a $\tau > 0$, we have that $-\tau^{\alpha/\alpha'} < -\tau$ (for $\tau < 1$, which is again be optimal here) has the same value of the numerator. From here, the claim is two-fold.

Claim 1: The optimal choice of τ for $\tau > 0$ has $\tau \geq \tau^*(\alpha)$.

To see this, consider that the first order condition is given by

$$-2\alpha\tau^{\alpha-1} - e^{-\tau^\alpha}\left[\frac{1}{\frac{\Gamma(1/\alpha')}{\alpha'}+\int_0^\tau e^{-x^\alpha}dx} - \frac{1}{\int_\tau^\infty e^{-x^\alpha}dx}\right].$$

[24]Of course, for $\tau > 1$ this is reversed, because the tails are smaller on the negative side of the distribution than the positive side of the distribution.

As $\frac{\Gamma(1/\alpha')}{\alpha'} > \frac{\Gamma(1/\alpha)}{\alpha}$, the first fraction is smaller, so that the positive component is larger. This means that for the first order condition to be satisfied for $\tau > 0$ we need $\tau > \tau^*(\alpha)$.

Claim 2: For all $\tau > \tau^*(\alpha)$, it is the case that $1 - F(\tau) > F(-\tau^{\alpha/\alpha'})$.

Notice that

$$1 - F(\tau) = \frac{1}{\frac{\Gamma(1/\alpha)}{\alpha} + \frac{\Gamma(1/\alpha')}{\alpha'}} \int_{\tau}^{\infty} e^{-x^{\alpha}} dx = \frac{1}{\frac{\Gamma(1/\alpha)}{\alpha} + \frac{\Gamma(1/\alpha')}{\alpha'}} \cdot \frac{\Gamma\left(\frac{1}{\alpha}, \tau^{\alpha}\right)}{\alpha}$$

and

$$F(-\tau^{\alpha/\alpha'}) = \frac{1}{\frac{\Gamma(1/\alpha)}{\alpha} + \frac{\Gamma(1/\alpha')}{\alpha'}} \int_{\tau^{\alpha/\alpha'}}^{\infty} e^{-x^{\alpha'}} dx = \frac{1}{\frac{\Gamma(1/\alpha)}{\alpha} + \frac{\Gamma(1/\alpha')}{\alpha'}} \cdot \frac{\Gamma\left(\frac{1}{\alpha'}, \tau^{\alpha}\right)}{\alpha'}$$

From here it is sufficient to show that $\frac{\Gamma(1/\alpha, \tau^*(\alpha)^{\alpha})}{\alpha} > \frac{\Gamma(1/\alpha', \tau^*(\alpha)^{\alpha})}{\alpha'}$ for all $\alpha' > \alpha$. Notice that the value of τ at which this is evaluated is important: it does not hold for all τ (indeed, it clearly does not hold for $\tau = 0$).

Subclaim: $\frac{\Gamma(1/\alpha, \tau^{\alpha})}{\alpha} > \frac{\Gamma(1/\alpha' \tau^{\alpha})}{\alpha'}$ for $\alpha \geq \alpha^*$ and $\tau^{\alpha} \geq \tau^*(\alpha^*)^{\alpha^*} > 0.001$.

Now, do the standard change of variable for $\beta := 1/\alpha$ and $x := \tau^{\alpha}$. I show that for $\beta < 1/\alpha^*$ it is the case that

$$\frac{\partial}{\partial \beta} \left(\Gamma(\beta, x) \cdot \beta \right) > 0$$

for all $x > 0.001$. The above is equivalent to (after some straightforward algebra in the same vein as above)

$$-\frac{1}{\beta} < \frac{\int_x^{\infty} \log(y) d\lambda_{\beta}(y)}{\int_x^{\infty} d\lambda_{\beta}(y)} \tag{14}$$

where the $d\lambda_{\beta}(y)$ notation is the same as in Lemma A.6. Notice first that the right-hand side of (14) is (i) increasing in x and (ii) increasing in β. Property (i) holds because is because log is increasing, and so by increasing x the average log value increases. Property (ii) follows from a similar logic: smaller β put more weight on smaller values of y (and hence smaller values of log).

This means that if (14) holds for $x^* := 0.001$, then the (sub)claim will be proved. From here, several straightforward computations prove the claim:

$$-2.16 < \frac{\int_{x^*}^{\infty} \log(y) d\lambda_{1/2.7}(y)}{\int_{x^*}^{\infty} d\lambda_{1/2.7}(y)},$$

dealing with $\beta \in \left(\frac{1}{2.7}, \frac{1}{\alpha^*}\right]$. Then,

$$-2.7 < \frac{\int_{x^*}^{\infty} \log(y) d\lambda_{1/4}(y)}{\int_{x^*}^{\infty} d\lambda_{1/4}(y)},$$

dealing with then $\beta \in \left(\frac{1}{4}, \frac{1}{\alpha^*}\right]$. Finally, it is the case that

$$-4 < \frac{\int_{x^*}^{\infty} \log(y) d\lambda_0(y)}{\int_{x^*}^{\infty} d\lambda_0(y)}.$$

This means that for all $\beta \in \left(0, \frac{1}{\alpha^*}\right]$ it is the case that (14) holds. This proves Claim 2.

From here, the conclusion is reached by simply putting together Claims 1 and 2: for any possible optimal positive τ, there is a negative threshold that performs strictly better, so the optimal threshold must be negative. $\square$

A.3 Proofs for Section 5.1

Proof of Lemma 3. This is a straightforward application of Theorem 1. $\square$

Proof of Theorem 3. The statement holds for general functions $p : \mathbb{N} \times \mathbb{R} \to [0, 1]$, apart from the final representation given in (4). The object of interest is

$$\lim_{\sigma \to \infty} \frac{\rho(\sigma; r, p)}{\rho(\sigma; r_f, p)}.$$

In order to simplify notation, I consider an equivalent limit in terms of μ. When signals are normalized, $f_L(s) = f\left(s + \frac{\mu}{\sigma}\right)$. This means that taking the limit as $\sigma \to \infty$ holding μ fixed is equivalent to taking the limit as $\mu \to 0$ while holding σ fixed. The notation is easier holding σ fixed, and so I consider everything in terms of the limit as $\mu \to 0$. Since the fixed value of σ is irrelevant under this notation, I set

$\sigma = 1$. First, define

$$f(s,\mu,\lambda) := f(s+\mu)^{\lambda} f(s-\mu)^{1-\lambda}.$$

Let $\rho(\mu; r_f, p, \lambda)$ denote the value of $\rho(\mu; r_f, p)$ evaluated at a fixed (but not necessarily optimal) λ. That is, $\rho(\mu; r_f, p) = \max_{\lambda\in[0,1]} \rho(\mu; r_f, p, \lambda)$. It can easily be seen that $\rho(\mu; r_f, p), \rho(\mu; r, p) \to 0$. Hence, l'Hopital's rule must be applied. As is shown below, it is also the case that $\frac{\partial}{\partial\mu}\rho(\mu; r_f, p), \frac{\partial}{\partial\mu}\rho(\mu; r_f, p) \to 0$. Hence, both the first and second derivative of the numerator and denominator must be computed.

Step 1: Denominator r_f. Assumption 1 guarantees that, for a fixed λ, the first derivative is given by

$$\begin{aligned}
&-\frac{\partial}{\partial\mu}\rho(\mu; r_f, p, \lambda) = \\
&\quad \int \left(\lambda \frac{f'(s+\mu)}{f(s+\mu)} - (1-\lambda)\frac{f'(s-\mu)}{f(s-\mu)}\right) f(s,\mu,\lambda) p(r_f, s) ds \\
&\quad +\lambda \frac{\int \left((1-p(r_f,s)) f(s+\mu)\right)^{\lambda-1} ds}{\int \left((1-p(r_f,s)) f(s-\mu)\right)^{\lambda-1} ds} \int \left((1-p(r_f,s)) f'(s+\mu)\right) ds \\
&\quad -(1-\lambda) \frac{\int \left((1-p(r_f,s)) f(s+\mu)\right)^{\lambda} ds}{\int \left((1-p(r_f,s)) f(s-\mu)\right)^{\lambda} ds} \int \left((1-p(r_f,s)) f'(s-\mu)\right) ds.
\end{aligned}$$

Notice that because of the envelope theorem, the derivative with respect to λ is equal to zero. Consider the limit as $\mu \to 0$. First, $f(s,\mu,\lambda) \to f(s)$. Hence, the limit is equal to

$$-\lim_{\mu\to 0} \frac{\partial}{\partial\mu}\rho(\mu; r_f, p, \lambda) = (2\lambda - 1)\int p(r_f, s)\ell_f(s) f(s) ds + (2\lambda-1)\int (1-p(r_f,s)) f'(s) ds.$$

This is equal to zero since $\ell_f(s) f(s) = f'(s)$, and $\int f'(s) ds = 0$ (because the expectation of the linear score function is zero). This means that the second derivative must be considered.[25] Again, the derivative with respect to λ is zero. This means that λ may be fixed, and the optimal value of λ derived at the end of the computation.

To simplify the null review notation, let $H_+(\mu, n) := \int (1-p(r_f, s)) f(s+\mu) ds$

[25] This is reliant on the numerator converging to zero at the same rate. I show this below.

(where n refers to "null"), and similarly $H_-(\mu, n)$. In a similar vein, let $H'_+(\mu, n) := \int(1 - p(r_f, s))f'(s + \mu)ds$. Similarly define $H'_-(\mu, n)$ (and $H''_\pm(\mu, n)$). Computing,

$$
\begin{aligned}
&-\frac{\partial^2}{\partial\mu^2}\rho(\mu; r_f, p, \lambda) = \\
&\quad \int \lambda(\lambda-1)\left(\left(\frac{f'(s+\mu)}{f(s+\mu)}\right)^2 + \left(\frac{f'(s+\mu)f'(s-\mu)}{f(s+\mu)f(s-\mu)}\right)\right) f(s,\mu,\lambda)p(r_f,s)ds \\
&\quad - \int \lambda(1-\lambda)\left(\left(\frac{f'(s-\mu)}{f(s-\mu)}\right)^2 + \left(\frac{f'(s-\mu)f'(s+\mu)}{f(s-\mu)f(s+\mu)}\right)\right) f(s,\mu,\lambda)p(r_f,s)ds \\
&\quad + \int \left(\lambda\frac{f''(s+\mu)}{f(s+\mu)} + (1-\lambda)\frac{f''(s-\mu)}{f(s-\mu)}\right) f(s,\mu,\lambda)p(r_f,s)ds \\
&\quad + \lambda(\lambda-1)\frac{H_+(\mu,n)^{\lambda-1}}{H_-(\mu,n)^{\lambda-1}}\left(\frac{H'_+(\mu,n)}{H_+(\mu,n)} + \frac{H'_-(\mu,n)}{H_-(\mu,n)}\right) H'_+(\mu,n) \\
&\quad - (1-\lambda)\lambda\frac{H_+(\mu,n)^{\lambda}}{H_-(\mu,n)^{\lambda}}\left(\frac{H'_+(\mu,n)}{H_+(\mu,n)} + \frac{H'_-(\mu,n)}{H_-(\mu,n)}\right) H'_-(\mu,n) \\
&\quad + \lambda\frac{H_+(\mu,n)^{\lambda-1}}{H_-(\mu,n)^{\lambda-1}} H''_+(\mu,n) + (1-\lambda)\frac{H_+(\mu,n)^{\lambda}}{H_-(\mu,n)^{\lambda}} H''_-(\mu,n).
\end{aligned}
$$

The limit is given by

$$
\begin{aligned}
&\lim_{\mu\to 0}\frac{\partial^2}{\partial\mu^2}\rho(\mu; r_f, p, \lambda) = \\
&\qquad 4\lambda(1-\lambda)\int\left(\frac{f'(s)}{f(s)}\right)^2 f(s)p(r_f,s) + \int f''(s)p(r_f,s)ds \\
&\qquad + 4\lambda(1-\lambda)\frac{H'_+(0,n)^2}{H_+(0,n)} + H''_+(0,n) \\
&= \quad 4\lambda(1-\lambda)\int\left(\frac{f'(s)}{f(s)}\right)^2 f(s)p(r_f,s) + \int p(r_f,s)f''(s)ds \\
&\qquad + 4\lambda(1-\lambda)\frac{\left(\int(1-p(r_f,s))f'(s)ds\right)^2}{\int(1-p(r_f,s))f(s)ds} + \int(1-p(r_f,s))f''(s)ds.
\end{aligned}
$$

As the anti-derivative of $f''(s)$ is $f'(s)$, it is the case that $\int f''(s)ds = 0$. Similarly $\int f'(s)ds = 0$, and $\int f(s)ds = 1$. This shows that this value is equal to the denominator in the statement of the proposition when $\lambda = \frac{1}{2}$. As $\lambda = \frac{1}{2}$ maximizes $\lambda(1-\lambda)$, it is the maximizer of this object, and so is the optimal λ.

This completes the derivations for the denominator. I now turn attention to the numerator. The computations are similar to those for null reviews above.

Step 2: Numerator r. The terms in the numerator are of the following form. Define

$$g_{\tilde{r}}(\mu, \lambda) := \left(\int_{\{s:r(s)\in\tilde{r}\}} f(s+\mu)p(r,s)ds\right)^{\lambda}\left(\int_{\{s:r(s)\in\tilde{r}\}} f(s-\mu)p(r,s)ds\right)^{1-\lambda}.$$

Similarly, for null reviews define

$$g_n(\mu, \lambda) := \left(\int f(s+\mu)(1-p(r,s))ds\right)^{\lambda}\left(\int f(s-\mu)(1-p(r,s))ds\right)^{1-\lambda}.$$

The value of the numerator for a fixed λ is of the form

$$\rho(\mu; r, p, \lambda) := 1 - \sum_{\tilde{r}\in\mathcal{R}_r} g_{\tilde{r}}(\mu, \lambda) - g_n(\mu, \lambda).$$

In order to make computations easier to follow, I compute the limits term by term. Notice that the computations for $g_n(\mu, \lambda)$ have already been computed above. For notational clarity, define

$$H_+(\mu, \tilde{r}) = \int_{\{r(s)\in\tilde{r}\}} f(s+\mu)p(r,s)ds.$$

and similarly all of the related functions $(H_-(\mu,\tilde{r}), H'_\pm(\mu,\tilde{r}), H''_\pm(\mu,\tilde{r}))$. Notice here that the term is $p(r,s)$ and not $1-p(r,s)$, because these are submitted reviews. Observe that

$$\frac{\partial}{\partial\mu} g_{\tilde{r}}(\mu,\lambda) = \lambda\frac{H_+(\mu,\tilde{r})^{\lambda-1}}{H_-(\mu,\tilde{r})^{\lambda-1}}H'_+(\mu,\tilde{r}) - (1-\lambda)\frac{H_+(\mu,\tilde{r})^{\lambda}}{H_-(\mu,\tilde{r})^{\lambda}}H'_-(\mu,\tilde{r}).$$

This means that

$$\lim_{\mu\to 0}\frac{\partial}{\partial\mu} g_{\tilde{r}}(\mu,\lambda) = (2\lambda-1)H'(0,\tilde{r}).$$

Recall that we have already computed the dynamics of $g_n(\mu,\lambda)$ above. This means that, regardless of λ, we have that

$$\begin{aligned}
-\lim_{\mu\to 0}\frac{\partial}{\partial\mu}\rho(\mu;r,p,\lambda) =&(2\lambda-1)\left(\sum_{\tilde{r}\in\mathcal{R}_r} H_+'(0,\tilde{r}) + H_+'(0,n)\right)\\
=&(2\lambda-1)\left(\sum_{\tilde{r}\in\mathcal{R}_r}\int_{\{r(s)\in\tilde{r}\}} f'(s)p(r,s)ds + \int f'(s)(1-p(r,s))ds\right)\\
=&(2\lambda-1)\int f'(s)ds = 0.
\end{aligned}$$

As in the case of full-detail reviews, the variation in λ does not contribute and so λ may be considered point-wise. In particular, this means that the second derivative must be considered. Computing,

$$\begin{aligned}
\frac{\partial^2}{\partial\mu^2}g_{\tilde{r}}(\mu,\lambda) =&\lambda(\lambda-1)\frac{H_+(\mu,\tilde{r})^{\lambda-1}}{H_-(\mu,\tilde{r})^{\lambda-1}}\left(\frac{H_+'(\mu,\tilde{r})}{H_+(\mu,\tilde{r})}+\frac{H_-'(\mu,\tilde{r})}{H_-(\mu,\tilde{r})}\right)H_+'(\mu,\tilde{r})\\
&-(1-\lambda)\lambda\frac{H_+(\mu,\tilde{r})^{\lambda}}{H_-(\mu,\tilde{r})^{\lambda}}\left(\frac{H_+'(\mu,\tilde{r})}{H_+(\mu,\tilde{r})}+\frac{H_-'(\mu,\tilde{r})}{H_-(\mu,\tilde{r})}\right)H_-'(\mu,\tilde{r})\\
&+\lambda\frac{H_+(\mu,\tilde{r})^{\lambda-1}}{H_-(\mu,\tilde{r})^{\lambda-1}}H_+''(\mu,\tilde{r})+(1-\lambda)\frac{H_+(\mu,\tilde{r})^{\lambda}}{H_-(\mu,\tilde{r})^{\lambda}}H_-''(\mu,\tilde{r}).
\end{aligned}$$

Notice that these computations are similar to those used for $g_n(\mu,\lambda)$ in the computation of the numerator's second derivative. Note again that the variation in λ does not contribute. This means that

$$\lim_{\mu\to 0}\frac{\partial^2}{\partial\mu^2}g_{\tilde{r}}(\mu,\lambda) = 4\lambda(\lambda-1)\frac{H_+'(0,\tilde{r})^2}{H_+(0,\tilde{r})}+H_+''(0,\tilde{r}).$$

All that is left is to collect terms and notice again that $\int f''(s)ds = 0$:

$$\begin{aligned}
\lim_{\mu\to 0}\frac{\partial^2}{\partial\mu^2}\rho(\mu;r,p,\lambda) =&-\sum_{\tilde{r}\in\mathcal{R}_r}\left(4\lambda(\lambda-1)\frac{H_+'(0,\tilde{r})^2}{H_+(0,\tilde{r})}+H_+''(0,\tilde{r})\right)\\
&-4\lambda(\lambda-1)\frac{H_+'(0,n)^2}{H_+(0,n)}-H_+''(0,n)\\
=&4\lambda(1-\lambda)\left(\sum_{\tilde{r}}\frac{H_+'(0,\tilde{r})^2}{H_+(0,\tilde{r})}+\frac{H_+'(0,n)^2}{H_+(0,n)}\right).
\end{aligned}$$

This last equality uses the fact that $\sum_{\tilde{r}} H''_+(0,\tilde{r}) + H''_+(0,n) = 0$. Observe that $\lambda = \frac{1}{2}$ is again the maximizer. This finishes the computation for the numerator. Finally, putting this result together with the computations from the denominator above proves the claim, after the functional form representation for p is used. $\square$

B Relationship between the Platform and Consumers' Preference over Information Structures

In this section, I study the perspective of a consumer choosing which source of information to use when making their decision. In practice, many platforms elicit multiple types of information from reviewers; often one can leave a coarse rating and then supplement the rating with a free-text full-detail review.

In the work above I take the perspective of the platform ex-ante choosing what type of review to elicit. In that choice the platform internalizes the randomness in the review decisions of reviewers. If there are multiple types of reviews that have been collected, when it comes time for consumers to choose between them, this randomness has been removed. So, the comparison is not between the rates of acquisition, but directly between the number of reviews of each type of review.

The comparison from the side of consumers is more similar to asking how many coarsened reviews "equals" one review of another type. The distinction between the two is subtle, but as shown below, the removal of risk leads the consumer to require more coarsened reviews. However, as individual reviews become uninformative, this difference disappears and so the metric of learning loss from Lemma 2 applies again.

Lemma B.1. Fix a consumer with a (finite) decision problem, two review systems r, r', and a degree of distinction μ. Then there exists an N large such that for all $n_r, n_{r'} > N$, n_r reviews of type r is preferred to $n_{r'}$ reviews of type r' if

$$\frac{n_r}{n_{r'}} > \frac{\log(1-\nu(\sigma;r'))}{\log(1-\nu(\sigma;r))}.$$

Proof. Let (n_r, r) denote the statistical experiment that generates n_r conditionally independent signals of drawn according to r. Then, because each of the signals are independent,

$$\begin{aligned}\nu(\sigma;(n_r,r)) =& 1-\min_{\lambda\in[0,1]}\int_{\mathcal{R}_r}\cdots\int_{\mathcal{R}_r}\prod_{i=1}^{n_r}\left(\gamma_L^r(d\tilde{r}_i)\right)^{\lambda}\left(\gamma_H^r(d\tilde{r}_i)\right)^{1-\lambda}\\ =&1-\min_{\lambda\in[0,1]}\prod_{i=1}^{n_r}\int_{\mathcal{R}_r}\left(\gamma_L^r(d\tilde{r}_i)\right)^{\lambda}\left(\gamma_H^r(d\tilde{r}_i)\right)^{1-\lambda}\\ =&1-(1-\nu(\sigma;r))^{n_r}.\end{aligned}$$

From here, the result comes from an application of either Torgersen's theorem or Theorem 1. Note that this result follows the same logic as Proposition 3 in Mu et al. (2021). See also Chernoff (1952), which discusses this relationship at some length. □

This condition is similar to, but not exactly, the condition of Theorem 1. As the next Proposition highlights, because the information for consumers is not random, consumers place more weight on "good" information than the platform. However, as information becomes very noisy, this difference shrinks and eventually disappears in the limit. Specifically, the relationship is as follows.

Proposition B.1. Let $\kappa(\sigma;r',r):=\frac{\nu(\sigma;r')}{\nu(\sigma;r)}$ and let $\zeta(\sigma;r',r):=\frac{\log(1-\nu(\sigma;r'))}{\log(1-\nu(\sigma;r))}$. Then,

$$\kappa(\sigma;r',r)<\zeta(\sigma;r',r)\iff\nu(\sigma;r')>\nu(\sigma;r).$$

However,

$$\lim_{\sigma\to\infty}\kappa(\sigma;r',r)=\lim_{\sigma\to\infty}\zeta(\sigma;r',r).$$

Proof. For all $x,y\in(0,1)$,

$$\frac{\log(1-x)}{\log(1-y)}>\frac{x}{y}\iff x>y.$$

This shows the first claim. For the second, one can apply l'Hopital's rule to the limit of $\sigma\to\infty$ for

$$\zeta(\sigma;r',r)=\frac{\log(1-\nu(\sigma;r'))}{\log(1-\nu(\sigma;r))}.$$

and find that

$$\lim_{\sigma\to\infty}\zeta(\sigma;r',r)=\lim_{\sigma\to\infty}\frac{\frac{\frac{\partial}{\partial\sigma}\nu(\sigma;r')}{\nu(\sigma;r')}}{\frac{\frac{\partial}{\partial\sigma}\nu(\sigma;r)}{\nu(\sigma;r)}}=\lim_{\sigma\to\infty}\frac{\frac{\partial}{\partial\sigma}\nu(\sigma;r')}{\frac{\partial}{\partial\sigma}\nu(\sigma;r)}=\lim_{\sigma\to\infty}\eta(\mu;r',r).$$

This shows the second claim, completing the proof. □

The platform's choice between ratings systems must necessarily reflect the randomness with which it receives information. This means that, relative to a consumer who does not need to factor-in this randomness, the platform values more frequent, but uninformative, reviews. This result suggests why in practice platforms collect multiple sources of information: consumers trade-off between sources of information in different ways than do platforms, because of the timing of their choice between sources of information.